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General time interval BSDEs under the weak monotonicity condition and nonlinear decomposition for general g -supermartingales

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ABSTRACT

The main purpose of this paper is to prove an existence and uniqueness result for solutions of a multidimensional backward stochastic differential equation (BSDE) with a general time interval (including the deterministic and stochastic cases), where the generator g of the BSDE is weakly monotonic and of general growth in y , and Lipschitz continuous in z , both non-uniformly with respect to t . And, the corresponding comparison theorem for the solutions of one-dimensional BSDEs is provided. As applications, we establish a nonlinear Doob-Meyer's decomposition theorem for general continuous g -supermartingales under an additional assumption of the generator g . Some new problems in our setting arise naturally and are well overcome. These results generalize and improve some known works.

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1. Introduction and preliminaries

Let k and d be two given positive integers, $(\Omega, \mathcal{F}, \mathbf{P})$ a completed probability space carrying a standard d -dimensional Brownian motion $(B_t)_{t \geq 0}$, and $(\mathcal{F}_t)_{t \geq 0}$ the natural σ -algebra filtration generated by $(B_t)_{t \geq 0}$. We assume that $(\mathcal{F}_t)_{t \geq 0}$ is right-continuous and complete and that $\mathcal{F}_T = \mathcal{F}$ for a given terminal time T satisfying $0 \leq T \leq \infty$. In this paper, we consider multidimensional backward stochastic differential equations (BSDEs) of the following form:

$$y_t = \xi + \int_t^T g(s, y_s, z_s) ds + \int_t^T dV_s - \int_t^T z_s dB_s, \quad t \in [0, T], \quad (1)$$

where the terminal condition ξ is a $\mathcal{F}_T$ -measurable k -dimensional random vector; the generator $g(\omega, t, y, z) : \Omega \times [0, T] \times \mathbf{R}^k \times \mathbf{R}^{k \times d} \mapsto \mathbf{R}^k$ is $(\mathcal{F}_t)$ -progressively measurable for each (y, z) ; V is a given $(\mathcal{F}_t)$ -adapted $\mathbf{R}^k$ -valued continuous process with each component having a finite variation. The solution $(y_t, z_t)_{t \in [0, T]}$ is a pair of $(\mathcal{F}_t)$ -adapted processes. We usually denote BSDE (1) by BSDE $(\xi, T, g + dV)$.

In the case of $V \equiv 0$, Pardoux and Peng [20] initially introduced the nonlinear version of finite time interval multidimensional BSDEs and obtained an existence and uniqueness

result of the square integrable solutions when the generator g is uniformly Lipschitz continuous in both y and z . Since then, there has been a tremendous interest in relaxing conditions of BSDEs from different aspects in order to apply the corresponding existence and uniqueness results to other fields, such as weakening the Lipschitz condition on g (see [3,11,13,14,18] and references therein), improving the finite terminal time into the general case (see [5,10,19], etc.), and studying the solutions under non-square integrable parameters (see [2,8], etc.).

To our knowledge, El Karoui et al. [7] first proposed the notation of g -supersolution by virtue of BSDEs, where the generator g satisfies the uniformly Lipschitz condition in (y, z) and V is an increasing, $(\mathcal{F}_t)$ -adapted and RCLL process with $V_0 = 0$. A g -supersolution of a linear BSDE is actually a superhedging strategy. And g -supersolutions facilitate the study of reflected BSDEs, see Peng and Xu [23] and Klimsiak [12]. The former concluded that solving a reflected BSDE is equivalent to finding the smallest g -supersolution of the BSDE with a special constraint. In this spirit, Peng and Xu [24] also solved an American option pricing problem in an incomplete market. Furthermore, Peng [21] provided the notion of g -martingales and proved a nonlinear Doob-Meyer's decomposition theorem for g -supermartingales by constructing a monotonic limit theorem for g -supersolutions. On the one hand, g -martingale is a generalization of classical martingales, i.e. g -martingales become linear martingales when $g = 0$; on the other hand, g -(super)martingales have close connection with g -expectations/evaluations, filtration-consistent nonlinear expectations and dynamic risk measures. We refer the readers to Bayraktar and Yao [1], Coquet et al. [6], Ma and Yao [17] and Peng [22] for more detailed relationships among them, as well as some basic properties of g -(super)martingales.

Lin [15,16] considered respectively the nonlinear Doob-Meyer's decomposition of g -supermartingales with respect to a general filtration and with jumps. Ma and Yao [17] researched the nonlinear decomposition, optional sampling theorem and upcrossing inequality for g -supermartingales under the quadratic growth condition. Shi et al. [25] studied the nonlinear decomposition of continuous g -supermartingales with uniformly continuous generators. For the purpose of solving reflected BSDEs, Klimsiak [12] solved BSDE (1) of one-dimensional case where the condition required by g is monotonicity and general growth in y , and Lipschitz continuity in z . Recently, Fan and Jiang [9] investigated the existence and uniqueness for solutions of multidimensional BSDEs where the condition for g with respect to y is extended to a weak monotonicity condition. This condition unifies some existing classical conditions, such as the monotonicity condition, Mao's condition, Osgood's condition and Constantin's condition. Additionally, motivated by the classical martingales theory, Chen and Wang [5] improved the result of Pardoux and Peng [20] to the general time interval case and obtained the convergence theorem of g -martingales.

Inspired by above results, we will study BSDE (1) with a general time interval under the weak monotonicity condition. We first show an existence and uniqueness result for a solution of BSDE (1) with a general time interval ($0 \leq T \leq \infty$), where the generator g satisfies a weak monotonicity condition with a general growth condition in y , and a Lipschitz continuity condition in z , both non-uniformly with respect to t (see Theorem 6). We also improve this result to a stochastic time interval case, i.e. the terminal time is a stopping time valued in $[0, T]$ (see Theorem 12). And then we prove the corresponding comparison theorem for the solutions of one-dimensional BSDEs (see

Theorem 13). Finally, we establish a nonlinear Doob-Meyer's decomposition theorem for general continuous g -supermartingales under an additional assumption of the generator g (see Theorem 18).

Here, we would like to emphasize the following three things. The first one is that when proving the existence result, Pardoux [19] and Fan and Jiang [9] first truncated $g(\cdot, 0, 0)$ and made a convolution to obtain the local Lipschitz condition, and then truncated the variable y to attain the global Lipschitz condition. Briand et al. [2] used a similar method to deal with the more general growth condition. Yet we think that truncating directly $g(\cdot, y, z)$ in the whole domain is more preferable in order to dominate the growth of g and its gradient. Surprisingly, due to this change we drop some intermediate steps and simplify greatly the proof procedure (see the first and second steps in the proof of Proposition 10 for details). The second thing needed to mention is that the existence and uniqueness result of the stochastic time interval case is not trivial. It is because the right hand side of the weak monotonicity inequality is a nonlinear function, which leads to the invalidation of methods used by Pardoux [19] and others. Here the stochastic time interval case is treated by the aid of the general time interval case. The third things should be stressed is that in the proof of nonlinear Doob-Meyer's decomposition theorem for general g -supermartingales, some new difficulties arise naturally due to the general time intervals and the general growth condition required by g in y . First of all, those techniques used in Shi et al. [25] (especially the technique in their Proposition 11) fail to obtain the nonlinear decomposition. Secondly, the weak convergence method used in Peng [21] can not be adopted directly in our setting since the boundedness of the process sequence involved the generator part in the usual $\mathcal{M}^2$ space (see the definition below) does not hold any longer due to the fact that T may be infinite and g may have a general growth in y , which leads to a loss of the corresponding weak compactness. These difficulties are successfully overcome by studying the weak sequential compactness of the above sequence in a new process space and employing the stopping time technique to treat the general growth condition of g in y (see the third step in the proof of Theorem 18 for details).

In this paper, the Euclidean norm of a vector $y \in \mathbf{R}^k$ will be defined by $|y|$, $\langle x, y \rangle$ will represent the inner product of $x, y \in \mathbf{R}^k$; and for a $k \times d$ matrix z , we define $|z| := \sqrt{\text{Tr}(zz^*)}$, where z^* is the transpose of z . Let $L^2(\mathcal{F}_T; \mathbf{R}^k)$ (resp. $L^2(\mathcal{F}_\tau; \mathbf{R}^k)$), where τ is a $(\mathcal{F}_t)$ -stopping time valued in $[0, T]$ be the set of $\mathbf{R}^k$ -valued and $\mathcal{F}_T$ -measurable (resp. $\mathcal{F}_\tau$ -measurable) random variables ξ such that $\|\xi\|_{L^2}^2 := \mathbf{E}[|\xi|^2] < \infty$ and $\mathcal{V}^2(0, T; \mathbf{R}^k)$ (or $\mathcal{V}^2$) the set of $\mathbf{R}^k$ -valued and $(\mathcal{F}_t)$ -adapted continuous processes with each component having a finite variation such that $V_0 = 0$ and $\mathbf{E}[|V|_T^2] < \infty$. Here and hereafter, $|V|_T$ represents the total variation of V on the interval $[0, T]$, i.e. $|V|_T = \sqrt{\sum_{i=1}^k |V^i|_T^2}$, where $|V^i|_T$ is the total variation of the i th component V^i of V on $[0, T]$. For each $(V_t)_{t \in [0, T]} \in \mathcal{V}^2$, we denote by dV the random measure generated by its trajectories.

Let $\mathcal{S}^2(0, T; \mathbf{R}^k)$ (or $\mathcal{S}^2$) denote the set of $\mathbf{R}^k$ -valued, $(\mathcal{F}_t)$ -adapted continuous processes $(Y_t)_{t \in [0, T]}$ such that

$$\|Y\|_{\mathcal{S}^2} := \left(\mathbf{E} \left[\sup_{t \in [0, T]} |Y_t|^2 \right] \right)^{1/2} < \infty.$$

Then, $\|\cdot\|_{\mathcal{S}^2}$ is a norm. Moreover, let $\mathcal{M}^2(0, T; \mathbf{R}^{k \times d})$ (or $\mathcal{M}^2$) denote the set of (equivalent classes of) $(\mathcal{F}_t)$ -progressively measurable $\mathbf{R}^{k \times d}$ -valued processes $(Z_t)_{t \in [0, T]}$ such that

$$\|Z\|_{\mathcal{M}^2} := \left(\mathbf{E} \left[\int_0^T |Z_t|^2 dt \right] \right)^{1/2} < \infty.$$

Then $\mathcal{M}^2$ is a Banach space endowed with this norm.

Hereafter, for each $0 \leq r \leq t \leq T$ we set $X_{r,t}^* := \sup_{s \in [r, t]} |X_s|$ for a given stochastic process $(X_s)_{s \in [0, T]}$ taking values in $\mathbf{R}^k$. And when $r = 0$, we denote it simply by X_t^* . Let $\mathbb{S}$ denote the set of nondecreasing and concave continuous functions $\kappa(\cdot) : \mathbf{R}^+ \mapsto \mathbf{R}^+$ satisfying $\kappa(0) = 0$, $\kappa(x) > 0$ for $x > 0$ and $\int_0^+ \kappa^{-1}(x) dx = \infty$. Since κ increases at most linearly, we will always denote by $A > 0$ the linear growth constant, i.e. $\kappa(x) \leq Ax + A$ for each $x \in \mathbf{R}^+$.

We should claim that all equalities and inequalities between random elements in the whole paper will be understood to hold almost surely.

As mentioned above, we will only deal with the BSDE which is an equation of type (1), where $\xi \in L^2(\mathcal{F}_T; \mathbf{R}^k)$ and $(V_t)_{t \in [0, T]} \in \mathcal{V}^2(0, T; \mathbf{R}^k)$. Throughout this paper, we use the following definition.

Definition 1: A pair of processes $(y_t, z_t)_{t \in [0, T]}$ is called a solution of BSDE (1), if $(y_t, z_t)_{t \in [0, T]}$ belongs to $\mathcal{S}^2(0, T; \mathbf{R}^k) \times \mathcal{M}^2(0, T; \mathbf{R}^{k \times d})$ and satisfies BSDE (1).

The rest of this paper is organized as follows. Section 2 lists some lemmas which will be used later and establishes an important a priori estimate for solutions of BSDE (1). Section 3 is devoted to the existence and uniqueness result for a solution of BSDE (1). Section 4 gives a corresponding comparison theorem for the solutions of one-dimensional BSDEs. And Section 5 proves the nonlinear Doob-Meyer's decomposition theorem for general continuous g -supermartingales.

2. Lemmas and an a priori estimate

Some useful lemmas will be introduced in this section and a crucial a priori estimate for solutions of BSDE (1) will be established below. They will be applied repeatedly later. Lemma 2 is a general Gronwall's inequality which was proved in Lemma 2.2 of Fan and Jiang [8].

Lemma 2 (Gronwall's inequality): Let $0 \leq T \leq \infty$, $\alpha(\cdot) : [0, T] \mapsto \mathbf{R}^+$ be a decreasing function, $\beta(\cdot) : [0, T] \mapsto \mathbf{R}^+$ satisfy $\int_0^T \beta(t) dt < \infty$ and $h(\cdot) : [0, T] \mapsto \mathbf{R}^+$ be a continuous function with $\sup_{t \in [0, T]} h(t) < \infty$ such that

$$h(t) \leq \alpha(t) + \int_t^T \beta(s) h(s) ds, \quad t \in [0, T].$$

Then we have

$$h(t) \leq \alpha(t) e^{\int_t^T \beta(s) ds}, \quad t \in [0, T].$$

The following Lemma 3 is a general version of Bihari's inequality, which is a direct corollary of Lemma 5 in Fan et al. [10].

Lemma 3: [Bihari's inequality] Assume that $0 \leq t \leq T \leq \infty$, $\beta(\cdot) : [t, T] \mapsto \mathbf{R}^+$ satisfies $\int_t^T \beta(s) ds < \infty$, $h(\cdot) : [t, T] \mapsto \mathbf{R}^+$ satisfies $\sup_{r \in [t, T]} h(r) < \infty$ and

$$h(r) \leq \int_r^T \beta(s) \kappa(h(s)) ds, \quad r \in [t, T],$$

where $\kappa \in \mathbb{S}$. Then we have that $h(r) = 0$ for all $r \in [t, T]$.

Next we establish the a priori estimate for solutions of general time interval BSDE (1). In stating it, it is useful to introduce the following assumption on the generator g , where $0 \leq T \leq \infty$:

- (A) There exist two nonnegative functions $\mu(\cdot), \lambda(\cdot) : [0, T] \mapsto \mathbf{R}^+$ with $\int_0^T (\mu(t) + \lambda^2(t)) dt < \infty$ such that $d\mathbf{P} \times dt$ - a.e., for each $y \in \mathbf{R}^k$ and $z \in \mathbf{R}^{k \times d}$,

$$\langle y, g(t, y, z) \rangle \leq \mu(t) \kappa(|y|^2) + \lambda(t) |y| |z| + f_t |y|,$$

where the function $\kappa \in \mathbb{S}$, $(f_t)_{t \in [0, T]}$ is a nonnegative, real valued and $(\mathcal{F}_t)$ -progressively measurable process with $\mathbf{E}[(\int_0^T f_t dt)^2] < \infty$.

Proposition 4: Assume that $0 \leq T \leq \infty$, assumption (A) holds for g , and $(y_t, z_t)_{t \in [0, T]}$ is a solution of BSDE (1), then there exists a constant $c > 0$ such that for each $0 \leq u \leq t \leq T$,

$$\begin{aligned} \mathbf{E}[|y_{t,T}^*|^2 | \mathcal{F}_u] + \mathbf{E}\left[\int_t^T |z_s|^2 ds \middle| \mathcal{F}_u\right] &\leq ce^{\int_t^T \lambda^2(s) ds} \left\{ \mathbf{E}[|\xi|^2 | \mathcal{F}_u] + \mathbf{E}[|V_T|^2 | \mathcal{F}_u] \right. \\ &\quad \left. + \int_t^T \mu(s) \kappa(\mathbf{E}[|y_s|^2 | \mathcal{F}_u]) ds + \mathbf{E}\left[\left(\int_t^T f_s ds\right)^2 \middle| \mathcal{F}_u\right] \right\}. \end{aligned}$$

Proof: Applying Itô's formula to $|y_t|^2$ yields that for each $t \in [0, T]$,

$$|y_t|^2 + \int_t^T |z_s|^2 ds = |\xi|^2 + 2 \int_t^T \langle y_s, g(s, y_s, z_s) \rangle ds + 2 \int_t^T \langle y_s, dV_s \rangle - 2 \int_t^T \langle y_s, z_s dB_s \rangle. \quad (2)$$

The inner product term including g can be enlarged via assumption (A) as follows,

$$\begin{aligned} 2 \langle y_s, g(s, y_s, z_s) \rangle &\leq 2\mu(s) \kappa(|y_s|^2) + 2\lambda(s) |y_s| |z_s| + 2f_s |y_s|, \\ &\leq 2\mu(s) \kappa(|y_s|^2) + 2\lambda^2(s) |y_s|^2 + \frac{1}{2} |z_s|^2 + 2f_s |y_s|. \end{aligned} \quad (3)$$

It follows from the Burkholder–Davis–Gundy (BDG) inequality that $\{\int_0^t \langle y_s, z_s dB_s \rangle\}_{t \in [0, T]}$ is a uniformly integrable martingale. In fact, for each $0 \leq u \leq t \leq T$, there exists a positive constant b such that

$$\begin{aligned}
2\mathbf{E} \left[\sup_{r \in [t, T]} \left| \int_r^T \langle y_s, z_s \, dB_s \rangle \right| \middle| \mathcal{F}_u \right] &\leq 2b\mathbf{E} \left[y_{t, T}^* \cdot \left(\int_t^T |z_s|^2 \, ds \right)^{1/2} \middle| \mathcal{F}_u \right] \\
&\leq \frac{1}{4}\mathbf{E} [|y_{t, T}^*|^2 | \mathcal{F}_u] + 4b^2\mathbf{E} \left[\int_t^T |z_s|^2 \, ds \middle| \mathcal{F}_u \right]. \quad (4)
\end{aligned}$$

Thus, plugging (3) into (2) and taking conditional expectation with respect to $\mathcal{F}_u$ yield that, for each $0 \leq u \leq t \leq T$,

$$\frac{1}{2}\mathbf{E} \left[\int_t^T |z_s|^2 \, ds \middle| \mathcal{F}_u \right] \leq X_u^t + 2\mathbf{E} \left[\int_t^T \langle y_s, \, dV_s \rangle \middle| \mathcal{F}_u \right] + 2\mathbf{E} \left[\int_t^T f_s |y_s| \, ds \middle| \mathcal{F}_u \right], \quad (5)$$

where

$$X_u^t := \mathbf{E} [|\xi|^2 | \mathcal{F}_u] + 2\mathbf{E} \left[\int_t^T \mu(s) \kappa(|y_s|^2) \, ds \middle| \mathcal{F}_u \right] + 2 \int_t^T \lambda^2(s) \mathbf{E} [|y_s|^2 | \mathcal{F}_u] \, ds.$$

By virtue of $2\alpha\beta \leq \alpha^2/4 + 4\beta^2$, we know that for each $0 \leq u \leq t \leq T$,

$$2\mathbf{E} \left[\int_t^T f_s |y_s| \, ds \middle| \mathcal{F}_u \right] \leq \frac{1}{4}\mathbf{E} [|y_{t, T}^*|^2 | \mathcal{F}_u] + 4\mathbf{E} \left[\left(\int_t^T f_s \, ds \right)^2 \middle| \mathcal{F}_u \right], \quad (6)$$

$$2\mathbf{E} \left[\int_t^T \langle y_s, \, dV_s \rangle \middle| \mathcal{F}_u \right] \leq \frac{1}{4}\mathbf{E} [|y_{t, T}^*|^2 | \mathcal{F}_u] + 4\mathbf{E} [|V|_T^2 | \mathcal{F}_u]. \quad (7)$$

It then follows from (2)–(4) and (6)–(7) that for each $0 \leq u \leq t \leq T$,

$$\begin{aligned}
&\frac{1}{4}\mathbf{E} [|y_{t, T}^*|^2 | \mathcal{F}_u] + \frac{1}{2}\mathbf{E} \left[\int_t^T |z_s|^2 \, ds \middle| \mathcal{F}_u \right] \\
&\leq X_u^t + 4\mathbf{E} [|V|_T^2 | \mathcal{F}_u] + 4b^2\mathbf{E} \left[\int_t^T |z_s|^2 \, ds \middle| \mathcal{F}_u \right] + 4\mathbf{E} \left[\left(\int_t^T f_s \, ds \right)^2 \middle| \mathcal{F}_u \right].
\end{aligned}$$

From (5) we can get that for each $0 \leq u \leq t \leq T$,

$$\begin{aligned}
4b^2\mathbf{E} \left[\int_t^T |z_s|^2 \, ds \middle| \mathcal{F}_u \right] &\leq 8b^2X_u^t + \frac{1}{8}\mathbf{E} [|y_{t, T}^*|^2 | \mathcal{F}_u] + 2^{10}b^4\mathbf{E} [|V|_T^2 | \mathcal{F}_u] \\
&\quad + 2^{10}b^4\mathbf{E} \left[\left(\int_t^T f_s \, ds \right)^2 \middle| \mathcal{F}_u \right].
\end{aligned}$$

Thus, combining the previous two inequalities yields that for each $0 \leq u \leq t \leq T$,

$$\begin{aligned} & \frac{1}{8} \mathbf{E} \left[|y_{t,T}^*|^2 \middle| \mathcal{F}_u \right] + \frac{1}{2} \mathbf{E} \left[\int_t^T |z_s|^2 ds \middle| \mathcal{F}_u \right] \\ & \leq (1 + 8b^2) X_u^t + (4 + 2^{10} b^4) \mathbf{E} \left[|V_T^2| \middle| \mathcal{F}_u \right] + (4 + 2^{10} b^4) \mathbf{E} \left[\left(\int_t^T f_s ds \right)^2 \middle| \mathcal{F}_u \right]. \end{aligned}$$

Finally, by setting $h(t) := \mathbf{E} \left[|y_{t,T}^*|^2 \middle| \mathcal{F}_u \right] + \mathbf{E} \left[\int_t^T |z_s|^2 ds \middle| \mathcal{F}_u \right]$ and then using the definition of X_u^t , Lemma 2, Fubini's theorem, Jensen's inequality and the concavity of $\kappa(\cdot)$, the desired result follows. $\square$

Remark 5: From (3) and (6) we can observe that if there exists a constant $\alpha > 0$ such that $d\mathbf{P} \times dt - \text{a.e. } |y_t| \leq \alpha$, then $(f_t)_{t \in [0, T]}$ in assumption (A) only needs to be integrable, i.e. $\mathbf{E} \left[\int_0^T f_t dt \right] < \infty$, and the estimate formulation in Proposition 4 still holds with $\mathbf{E} \left[\int_t^T \alpha f_s ds \middle| \mathcal{F}_u \right]$ replacing $\mathbf{E} \left[\left(\int_t^T f_s ds \right)^2 \middle| \mathcal{F}_u \right]$.

3. Existence and uniqueness for solutions of multidimensional BSDEs

In this section, we first introduce the existence and uniqueness result and two examples. Then we demonstrate the detailed proof of this result. Finally, the stochastic time interval case of this result is provided.

3.1. Statement of the existence and uniqueness result

We first present the following assumptions with respect to the generator g of BSDE (1) in the case $0 \leq T \leq \infty$. In stating them we always suppose that $u(\cdot), v(\cdot) : [0, T] \mapsto \mathbf{R}^+$ are two deterministic functions with $\int_0^T (u(t) + v^2(t)) dt < \infty$.

- (H1) $\mathbf{E} \left[\left(\int_0^T |g(t, 0, 0)| dt \right)^2 \right] < \infty$;
- (H2) $d\mathbf{P} \times dt - \text{a.e.}$, for each $z \in \mathbf{R}^{k \times d}$, $y \mapsto g(t, y, z)$ is continuous;
- (H3) g has a general growth in y , i.e. for each $r \in \mathbf{R}^+$, we have

$$\psi_r(t) := \sup_{|y| \leq r} |g(t, y, 0) - g(t, 0, 0)| \in L^1([0, T] \times \Omega);$$

- (H4) g satisfies a weak monotonicity condition in y non-uniformly with respect to t , i.e. there exists a function $\rho \in \mathbb{S}$ such that $d\mathbf{P} \times dt - \text{a.e.}$, for each $y_1, y_2 \in \mathbf{R}^k$ and $z \in \mathbf{R}^{k \times d}$,

$$\langle y_1 - y_2, g(t, y_1, z) - g(t, y_2, z) \rangle \leq u(t) \rho(|y_1 - y_2|^2);$$

- (H5) g is Lipschitz continuous in z non-uniformly with respect to t , i.e. $d\mathbf{P} \times dt - \text{a.e.}$, for each $y \in \mathbf{R}^k$ and $z_1, z_2 \in \mathbf{R}^{k \times d}$, we have

$$|g(t, y, z_1) - g(t, y, z_2)| \leq v(t)|z_1 - z_2|.$$

The main result of this section is as follows.

Theorem 6: Assume that $0 \leq T \leq \infty$ and g satisfies assumptions (H1)–(H5). Then for each $\xi \in L^2(\mathcal{F}_T; \mathbf{R}^k)$ and $(V_t)_{t \in [0, T]} \in \mathcal{V}^2$, BSDE (1) admits a unique solution $(y_t, z_t)_{t \in [0, T]}$ in $\mathcal{S}^2 \times \mathcal{M}^2$.

Remark 7: It should be noted that in the corresponding assumptions of Theorem 2.1 in Fan and Jiang [9] and Theorem 3.8 in Klimsiak [12], the $u(t)$ and $v(t)$ appearing in (H4) and (H5) are assumed to be bounded by a constant $c > 0$ since they worked with continuous functions on a compact time interval. And, their results will be invalid in the case $T = \infty$. In our framework the only requirement of the integrability for $u(t)$ and $v(t)$ can avoid these two restrictions.

Example 8: Let $0 < T < \infty$, $k = 1$ and

$$g(t, y, z) = |\ln t|(-e^y + h(|y|)) + \frac{|z|}{\sqrt[4]{t}} + |B_t|,$$

where $h(x) := (-x \ln x) \mathbf{1}_{0 \leq x \leq \delta} + (h'(\delta -)(x - \delta) + h(\delta)) \mathbf{1}_{x > \delta}$ with δ small enough.

It is not difficult to check that g satisfies assumptions (H1)–(H5) with $u(t) = |\ln t|$, $v(t) = 1/\sqrt[4]{t}$ and $\rho(x) = h(x)$. It then follows from Theorem 6 that for each $\xi \in L^2(\mathcal{F}_T; \mathbf{R})$ and $V \in \mathcal{V}^2$, BSDE $(\xi, T, g + dV)$ admits a unique solution in $\mathcal{S}^2 \times \mathcal{M}^2$. But because $u(t)$ and $v(t)$ are unbounded, this conclusion can not be obtained by Theorem 2.1 in Fan and Jiang [9] and Theorem 3.8 in Klimsiak [12]. Remark 7 applies.

Example 9: Let $0 \leq T \leq \infty$, $k = 2$ and

$$g(t, y, z) = t^2 e^{-t} \left[\frac{e^{-y_1} + h(|y_2|)}{e^{-y_2} + h(|y_1|)} \right] + \frac{1}{\sqrt{1+t^2}} \left[\frac{|z_1|}{|z_2|} \right] + \frac{t^2}{t^4 + 1} |B_t|,$$

where y_i and z_i ($i = 1, 2$) respectively stand for the i th component of the vector y and the i th row of the matrix z , and $h(\cdot)$ is defined in Example 8. It is not hard to verify that g satisfies assumptions (H1)–(H5) with $u(t) = t^2 e^{-t}$, $v(t) = 1/\sqrt{1+t^2}$ and $\rho(x) = h(x)$. Thus, by Theorem 6 we know that for each $\xi \in L^2(\mathcal{F}_T; \mathbf{R}^2)$ and $V \in \mathcal{V}^2$, BSDE $(\xi, T, g + dV)$ admits a unique solution in $\mathcal{S}^2 \times \mathcal{M}^2$. This conclusion can not be obtained by Theorem 2.1 in Fan and Jiang [9] and Theorem 3.8 in Klimsiak [12] when $T = \infty$.

3.2. Proof of Theorem 6

Let us first prove the uniqueness part and then the existence part.

Proof of the uniqueness part of Theorem 6: Suppose that g satisfies assumptions (H1)–(H5), $\xi \in L^2(\mathcal{F}_T; \mathbf{R}^k)$ and $V \in \mathcal{V}^2$. As usual, let $(y_t^i, z_t^i)_{t \in [0, T]}$ ($i = 1, 2$) be two solutions of BSDE (1) and set $\hat{y} := y^1 - y^2$, $\hat{z} := z^1 - z^2$. Then $(\hat{y}_t, \hat{z}_t)_{t \in [0, T]}$ is a solution of the following BSDE:

$$\hat{y}_t = \int_t^T \hat{g}(s, \hat{y}_s, \hat{z}_s) ds - \int_t^T \hat{z}_s dB_s, \quad t \in [0, T], \quad (8)$$

where $\hat{g}(t, y, z) := g(t, y + y_t^2, z + z_t^2) - g(t, y_t^2, z_t^2)$ for each $(y, z) \in \mathbf{R}^k \times \mathbf{R}^{k \times d}$. It follows from (H4) and (H5) that $d\mathbf{P} \times dt - \text{a.e.}$, for each $(y, z) \in \mathbf{R}^k \times \mathbf{R}^{k \times d}$,

$$\begin{aligned} \langle y, \hat{g}(t, y, z) \rangle &= \langle y, g(t, y + y_t^2, z + z_t^2) - g(t, y_t^2, z + z_t^2) \rangle \\ &\quad + \langle y, g(t, y_t^2, z + z_t^2) - g(t, y_t^2, z_t^2) \rangle \\ &\leq u(t)\rho(|y|^2) + v(t)|y||z|, \end{aligned}$$

which means that assumption (A) is satisfied by $\hat{g}(t, y, z)$ of BSDE (8) with $\kappa(\cdot) = \rho(\cdot)$, $\mu(\cdot) = u(\cdot)$, $\lambda(\cdot) = v(\cdot)$ and $f_t \equiv 0$. Thus, it follows from Proposition 4 with $u = 0$ that there exists a constant $c > 0$ such that for each $t \in [0, T]$,

$$\mathbf{E} [|\hat{y}_{t,T}^*|^2] + \mathbf{E} \left[\int_t^T |\hat{z}_s|^2 ds \right] \leq ce^{\int_0^T v^2(s) ds} \int_t^T u(s) \rho(\mathbf{E} [|\hat{y}_{s,T}^*|^2]) ds.$$

Bihari's inequality (see Lemma 3) yields that for each $t \in [0, T]$,

$$\mathbf{E} [|\hat{y}_{t,T}^*|^2] + \mathbf{E} \left[\int_t^T |\hat{z}_s|^2 ds \right] = 0.$$

The proof of the uniqueness part is then complete. $\square$

Next we will prove the existence part of Theorem 6. The proof method is partly aspired by Fan and Jiang [9]. In our proof procedure, the local Lipschitz condition is not employed after convolution because we truncate the generator g in the whole domain. Besides, when dealing with the general growth condition in y we do not use another stronger growth condition as auxiliary. Hence, our proofs are simplified greatly.

We first prove the case that g is independent of z , see the following Proposition 10.

Proposition 10: Assume that $0 \leq T \leq \infty$, g is independent of z and satisfies (H1)–(H4). Then for each $\xi \in L^2(\mathcal{F}_T; \mathbf{R}^k)$ and $(V_t)_{t \in [0, T]} \in \mathcal{V}^2$, BSDE (1) admits a solution $(y_t, z_t)_{t \in [0, T]}$ in $\mathcal{S}^2 \times \mathcal{M}^2$.

Proof: Let $\xi \in L^2(\mathcal{F}_T; \mathbf{R}^k)$, $V_t \in \mathcal{V}^2$, g be independent of z and satisfy (H1)–(H4), which are rewritten as the following type for readers' convenience:

- (H1') $\mathbf{E} \left[\left(\int_0^T |g(t, 0)| dt \right)^2 \right] < \infty$;
- (H2') $d\mathbf{P} \times dt - \text{a.e.}$, $y \mapsto g(t, y)$ is continuous;
- (H3') for each $r \in \mathbf{R}^+$, we have $\psi_r(t) := \sup_{|y| \leq r} |g(t, y) - g(t, 0)| \in L^1([0, T] \times \Omega)$;
- (H4') there exists a function $\rho \in \mathbb{S}$ such that $d\mathbf{P} \times dt - \text{a.e.}$, for each $y_1, y_2 \in \mathbf{R}^k$ and $z \in \mathbf{R}^{k \times d}$,

$$\langle y_1 - y_2, g(t, y_1) - g(t, y_2) \rangle \leq u(t)\rho(|y_1 - y_2|^2).$$

The proof will be divided into three steps.

First Step: In this step, with the help of Proposition 4, by applying an approximation method via convolution smoothing inspired by Fan and Jiang [9], we will prove the existence of a solution in $\mathcal{S}^2 \times \mathcal{M}^2$ for the following BSDE:

$$y_t = \xi + \int_t^T g(s, y_s) ds + \int_t^T dV_s - \int_t^T z_s dB_s, \quad t \in [0, T], \quad (9)$$

under assumptions (H2') and (H4'), provided that there exists a nonnegative constant K such that

$$|\xi| \leq K, \quad |V|_T \leq K \quad \text{and} \quad |g(t, y)| \leq Ke^{-t}, \quad \forall y \in \mathbf{R}^k, \quad d\mathbf{P} \times dt - \text{a.e.} \quad (10)$$

In fact, for each $n \geq 1$, let $\phi_n(x) := n^k \phi(nx)$, where $\phi(\cdot) : \mathbf{R}^k \mapsto \mathbf{R}^+$ is a nonnegative C^∞ function with the unit ball as compact support and which satisfies $\int_{\mathbf{R}^k} \phi(x) dx = 1$. We define for each $n \geq 1$ and $(\omega, t, y) \in \Omega \times [0, T] \times \mathbf{R}^k$,

$$g_n(t, y) := (\phi_n(\cdot) * g(t, \cdot))(y) = \int_{\mathbf{R}^k} \phi_n(x) g(t, y - x) dx = \int_{\mathbf{R}^k} \phi_n(y - x) g(t, x) dx.$$

Then g_n is a $(\mathcal{F}_t)$ -progressively measurable process, and for each $n \geq 1$ and $y \in \mathbf{R}^k$ we have

$$g_n(t, y) = \int_{\mathbf{R}^k} \phi(x) g\left(t, y - \frac{x}{n}\right) dx = \int_{\{x: |x| \leq 1\}} \phi(x) g\left(t, y - \frac{x}{n}\right) dx. \quad (11)$$

It is straightforward to verify that (H2') and (H4') are satisfied by $g_n(t, y)$ for each $n \geq 1$. Moreover, it follows from (10) that $d\mathbf{P} \times dt - \text{a.e.}$, for each $n \geq 1$ and $y \in \mathbf{R}^k$,

$$|\nabla g_n(t, y)| \leq Ke^{-t} \int_{\mathbf{R}^k} |\nabla \phi_n(y - x)| dx = Ke^{-t} \int_{\{z: |z| \leq 1/n\}} |\nabla \phi_n(z)| dz \leq K'_n e^{-t},$$

where $K'_n > 0$ is a constant depending only on n and K . Thus, we have that $d\mathbf{P} \times dt - \text{a.e.}$, for each $n \geq 1$ and $y_1, y_2 \in \mathbf{R}^k$,

$$|g_n(t, y_1 - V_t) - g_n(t, y_2 - V_t)| \leq K'_n e^{-t} |y_1 - y_2|.$$

In addition, (10) gives that $d\mathbf{P} \times dt - \text{a.e.}$, for each $n \geq 1$,

$$|g_n(t, -V_t)| \leq \int_{\mathbf{R}^k} \phi_n(x) |g(t, -V_t - x)| dx \leq Ke^{-t},$$

and hence we get that

$$\mathbf{E} \left[\left(\int_0^T |g_n(t, -V_t)| dt \right)^2 \right] < \infty.$$

It then follows from Theorem 1.2 in Chen and Wang [5] for the case of $T \leq \infty$, an extension of the known existence and uniqueness result of Pardoux and Peng [20] for the case of $T < \infty$, that the following BSDE:

$$\bar{y}_t^n = \xi + V_T + \int_t^T g_n(s, \bar{y}_s^n - V_s) ds - \int_t^T \bar{z}_s^n dB_s$$

admits a unique solution $(\bar{y}_t^n, \bar{z}_t^n)_{t \in [0, T]}$ in $\mathcal{S}^2 \times \mathcal{M}^2$ for each $n \geq 1$. Thus, we can easily check that $(y_t^n, z_t^n)_{t \in [0, T]} := (\bar{y}_t^n - V_t, \bar{z}_t^n)_{t \in [0, T]}$ is just a solution of BSDE $(\xi, T, g_n + dV)$.

In the sequel, we shall show that $\{(y^n, z^n)\}_{n=1}^\infty$ is a Cauchy sequence in the space $\mathcal{S}^2 \times \mathcal{M}^2$. First, for each $n \geq 1$ and $y \in \mathbf{R}^k$, it follows from (10) and (11) that $d\mathbf{P} \times dt - \text{a.e.}$, $\langle y, g_n(t, y) \rangle \leq Ke^{-t} |y|$, which means that assumption (A) is satisfied by $g_n(t, y)$ with

$\mu(\cdot) \equiv 0, \kappa(\cdot) \equiv 0, \lambda(\cdot) \equiv 0, f_t = Ke^{-t}$. Hence, it follows from Proposition 4 and (10) that there exists a constant $C > 0$ such that for each $n \geq 1$ and $0 \leq u = t \leq T$,

$$|y_t^n|^2 + \mathbf{E} \left[\int_t^T |z_s^n|^2 ds \middle| \mathcal{F}_t \right] \leq C \mathbf{E} \left[|\xi|^2 + |V_T|^2 + \left(\int_0^T Ke^{-t} dt \right)^2 \middle| \mathcal{F}_t \right] \leq 3CK^2. \quad (12)$$

On the other hand, for each $n \geq 1$ and $m \geq 1$, let $\hat{y}^{n,m} := y^n - y^m, \hat{z}^{n,m} := z^n - z^m$. Then we have

$$\hat{y}_t^{n,m} = \int_t^T \hat{g}^{n,m}(s, \hat{y}_s^{n,m}) ds - \int_t^T \hat{z}_s^{n,m} dB_s, \quad t \in [0, T],$$

where $\hat{g}^{n,m}(t, y) := g_n(t, y + y_t^m) - g_m(t, y_t^m)$ for each $y \in \mathbf{R}^k$. It follows from (H4') for g_n that $d\mathbf{P} \times dt - \text{a.e.}$, for each $y \in \mathbf{R}^k$,

$$\langle y, \hat{g}^{n,m}(t, y) \rangle \leq u(t)\rho(|y|^2) + |y||g_n(t, y_t^m) - g_m(t, y_t^m)|.$$

Then assumption (A) is satisfied by $\hat{g}^{n,m}(t, y)$ with $\mu(\cdot) = u(\cdot), \kappa(\cdot) = \rho(\cdot), \lambda(\cdot) \equiv 0$ and $f_t = |g_n(t, y_t^m) - g_m(t, y_t^m)|$. Thus, Proposition 4 with $u = 0$ indicates that there exists a constant $\bar{C} > 0$ such that for each $t \in [0, T]$,

$$\begin{aligned} & \mathbf{E} \left[|\hat{y}_{t,T}^{n,m,*}|^2 \right] + \mathbf{E} \left[\int_t^T |\hat{z}_s^{n,m}|^2 ds \right] \\ & \leq \bar{C} \int_t^T u(s)\rho \left(\mathbf{E} \left[|\hat{y}_{s,T}^{n,m,*}|^2 \right] \right) ds + \bar{C} \mathbf{E} \left[\left(\int_0^T |g_n(s, y_s^m) - g_m(s, y_s^m)| ds \right)^2 \right]. \end{aligned} \quad (13)$$

Furthermore, it follows from (11) that $d\mathbf{P} \times ds - \text{a.e.}$, for each $n, m \geq 1$,

$$|g_n(s, y_s^m) - g_m(s, y_s^m)| \leq \int_{\{x: |x| \leq 1\}} f^{n,m,s}(x) \phi(x) dx$$

with

$$f^{n,m,s}(x) := \left| g \left(s, y_s^m - \frac{x}{n} \right) - g \left(s, y_s^m - \frac{x}{m} \right) \right|.$$

By (H2'), (12) and the fact that a function which is continuous on a compact set is uniformly continuous on that set, we can conclude that $d\mathbf{P} \times ds - \text{a.e.}$, for each $x \in \mathbf{R}^k, f^{n,m,s}(x) \rightarrow 0$ as $n, m \rightarrow \infty$. And, it follows from (10) that $d\mathbf{P} \times ds - \text{a.e.}$, for each $n, m \geq 1$ and $x \in \mathbf{R}^k, |f^{n,m,s}(x)| \leq 2Ke^{-s}$. Thus, applying Lebesgue's dominated convergence theorem twice yields that

$$\lim_{n,m \rightarrow \infty} \mathbf{E} \left[\left(\int_0^T |g_n(s, y_s^m) - g_m(s, y_s^m)| ds \right)^2 \right] = 0. \quad (14)$$

Now we set, for each $t \in [0, T]$,

$$h^{n,m}(t) := \mathbf{E} \left[|\hat{y}_{t,T}^{n,m,*}|^2 \right] + \mathbf{E} \left[\int_t^T |\hat{z}_s^{n,m}|^2 ds \right] \quad \text{and} \quad h(t) := \limsup_{n,m \rightarrow \infty} h^{n,m}(t).$$

Then, $h(t)$ is finite by (12). By taking the upper limit with respect to n and m in (13), in view of Fatou's lemma, the monotonicity and continuity of $\rho(\cdot)$ and (14), we can deduce that for each $t \in [0, T]$, $h(t) \leq \overline{C} \int_t^T u(s) \rho(h(s)) ds$. Thus, Bihari's inequality (see Lemma 3) yields that

$$\lim_{n,m \rightarrow \infty} \left\{ \mathbf{E} [|\hat{y}_T^{n,m,*}|^2] + \mathbf{E} \left[\int_0^T |\hat{z}_s^{n,m}|^2 ds \right] \right\} = 0,$$

which means that $\{(y^n, z^n)\}_{n=1}^\infty$ is a Cauchy sequence in the space $\mathcal{S}^2 \times \mathcal{M}^2$. We denote the limit process of this Cauchy sequence in $\mathcal{S}^2 \times \mathcal{M}^2$ by $(y_t, z_t)_{t \in [0, T]}$, and pass to the limit under the uniform convergence in probability (ucp) for BSDE $(\xi, T, g_n + dV)$, in view of (10)–(12), (H2') and Lebesgue's dominated convergence theorem, to see that $(y_t, z_t)_{t \in [0, T]}$ solves BSDE (9).

Second Step: In this step, we shall prove that BSDE (9) has a solution in $\mathcal{S}^2 \times \mathcal{M}^2$ under assumptions (H2')–(H4') provided that there exists a nonnegative constant K such that

$$|\xi| \leq K, |V|_T \leq K \quad \text{and} \quad |g(t, 0)| \leq Ke^{-t}, \quad d\mathbf{P} \times dt - \text{a.e.} \quad (15)$$

Assume now that (H2')–(H4') and (15) hold true. For some positive real $r > 0$, which will be chosen later, let θ_r be a smooth function such that $0 \leq \theta_r(y) \leq 1$, $\theta_r(y) = 1$ for $|y| \leq r$ and $\theta_r(y) = 0$ as soon as $|y| \geq r + 1$. We define for each $n \geq 1$ and $y \in \mathbf{R}^k$,

$$g^n(t, y) := \theta_r(y)(g(t, y) - g(t, 0)) \frac{ne^{-t}}{\psi_{r+1}(t) \vee (ne^{-t})} + g(t, 0).$$

Then it follows from the definitions of $\psi_{r+1}(\cdot)$ and $\theta_r(\cdot)$ with (15) and (H3') that $d\mathbf{P} \times dt - \text{a.e.}$, for each $n \geq 1$ and $y \in \mathbf{R}^k$,

$$|g^n(t, y)| \leq (n + K)e^{-t}.$$

Clearly, g^n satisfies (H2'). We can also deduce that assumption (H4') remains valid for $g^n(t, y)$. Indeed, for each y_1 and $y_2 \in \mathbf{R}^k$, if $|y_1| > r + 1$ and $|y_2| > r + 1$, (H4') is trivially satisfied and thus we reduce to the case where $|y_2| \leq r + 1$. By adding and subtracting $\theta_r(y_1)g(t, y_2)$ we can derive that $d\mathbf{P} \times dt - \text{a.e.}$, for each $n \geq 1$,

$$\begin{aligned} \langle y_1 - y_2, g^n(t, y_1) - g^n(t, y_2) \rangle &= \frac{ne^{-t}}{\psi_{r+1}(t) \vee (ne^{-t})} \theta_r(y_1) \langle y_1 - y_2, g(t, y_1) - g(t, y_2) \rangle \\ &\quad + \frac{ne^{-t}}{\psi_{r+1}(t) \vee (ne^{-t})} (\theta_r(y_1) - \theta_r(y_2)) \\ &\quad \times \langle y_1 - y_2, g(t, y_2) - g(t, 0) \rangle. \end{aligned}$$

It follows from (H4') for g that the first term on the right hand side is less than $u(t)\rho(|y_1 - y_2|^2)$. For the second term, since θ_r is c_r -Lipschitz and $|y_2| < r + 1$, we can get that, in view of (H3'), it is less than $nc_re^{-t}|y_1 - y_2|^2$. Hence, we have that $d\mathbf{P} \times dt - \text{a.e.}$, for each $n \geq 1$ and $y_1, y_2 \in \mathbf{R}^k$,

$$\langle y_1 - y_2, g^n(t, y_1) - g^n(t, y_2) \rangle \leq \bar{u}_n(t) \bar{\rho}(|y_1 - y_2|^2),$$

where $\bar{u}_n(t) := u(t) + nc_re^{-t}$ for each $t \in [0, T]$ and $n \geq 1$, and $\bar{\rho}(x) := \rho(x) + x$ for each $x \in \mathbf{R}^+$. It is obvious that $\int_0^T \bar{u}_n(t) dt < \infty$ for each $n \geq 1$, and that $\bar{\rho}(\cdot)$ is concave, nondecreasing and $\bar{\rho}(0) = 0$, $\bar{\rho}(x) > 0$ for each $x > 0$. And since $\rho(\cdot)$ is concave we can deduce that for each $x \in [0, 1]$,

$$\rho(x) = \rho(x \cdot 1 + (1-x) \cdot 0) \geq x\rho(1) + (1-x)\rho(0) = x\rho(1),$$

which yields that

$$\int_{0+} \frac{dx}{\bar{\rho}(x)} = \int_{0+} \frac{dx}{\rho(x) + x} \geq \frac{\rho(1)}{\rho(1) + 1} \int_{0+} \frac{dx}{\rho(x)} = \infty.$$

Now, we have proved that g^n fulfills all the assumptions required in the first step. It then follows that BSDE $(\xi, T, g^n + dV)$ admits a solution $(y_t^n, z_t^n)_{t \in [0, T]}$ in the space $\mathcal{S}^2 \times \mathcal{M}^2$.

Moreover, by assumption (H4') and (15) we know that $d\mathbf{P} \times dt$ - a.e., for each $y \in \mathbf{R}^k$,

$$\begin{aligned} \langle y, g^n(t, y) \rangle &= \theta_r(y) \frac{ne^{-t}}{\psi_{r+1}(t) \vee (ne^{-t})} \langle y, g(t, y) - g(t, 0) \rangle + \langle y, g(t, 0) \rangle \\ &\leq u(t)\rho(|y|^2) + Ke^{-t}|y|, \end{aligned}$$

which means that assumption (A) holds true for $g^n(t, y)$ with $\mu(\cdot) = u(\cdot)$, $\kappa(\cdot) = \rho(\cdot)$, $\lambda(\cdot) \equiv 0$ and $f_t = Ke^{-t}$. Then it follows from Proposition 4 and (15) that there exists a constant $\tilde{C} > 0$ such that for each $n \geq 1$ and $0 \leq u \leq t \leq T$,

$$\mathbf{E}[|y_t^n|^2 | \mathcal{F}_u] + \mathbf{E}\left[\int_t^T |z_s^n|^2 ds \middle| \mathcal{F}_u\right] \leq \tilde{C} \int_t^T u(s)\rho(\mathbf{E}[|y_s^n|^2 | \mathcal{F}_u]) ds + \tilde{C}K^2.$$

Given that $\rho(x) \leq Ax + A$ for each $x \in \mathbf{R}^+$, Gronwall's inequality (see Lemma 2) yields that for each $n \geq 1$ and $0 \leq u \leq t \leq T$,

$$\mathbf{E}[|y_t^n|^2 | \mathcal{F}_u] + \mathbf{E}\left[\int_t^T |z_s^n|^2 ds \middle| \mathcal{F}_u\right] \leq \left(\tilde{C}K^2 + A\tilde{C} \int_0^T u(s) ds\right) e^{A\tilde{C} \int_0^T u(s) ds}.$$

Now we choose a constant r such that r^2 equals to the right hand side term of the previous inequality. Then for each $n \geq 1$,

$$d\mathbf{P} \times dt \text{ - a.e., } |y_t^n| \leq r \text{ and } \mathbf{E}\left[\int_0^T |z_s^n|^2 ds\right] \leq r^2. \quad (16)$$

Hence, $(y_t^n, z_t^n)_{t \in [0, T]}$ is a solution of BSDE $(\xi, T, \tilde{g}^n + dV)$, where

$$\tilde{g}^n(t, y) := (g(t, y) - g(t, 0)) \frac{ne^{-t}}{\psi_{r+1}(t) \vee (ne^{-t})} + g(t, 0).$$

In the sequel, we show $\{(y^n, z^n)\}_{n=1}^\infty$ is a Cauchy sequence in $\mathcal{S}^2 \times \mathcal{M}^2$. For each $n, i \geq 1$, we set $\hat{y}^{n,i} := y^{n+i} - y^n$ and $\hat{z}^{n,i} := z^{n+i} - z^n$. Then

$$\hat{y}_t^{n,i} = \int_t^T \tilde{g}^{n,i}(s, \hat{y}^{n,i}) ds - \int_t^T \hat{z}_s^{n,i} dB_s,$$

where $\tilde{g}^{n,i}(t, y) := \tilde{g}^{n+i}(t, y + y_t^n) - \tilde{g}^n(t, y_t^n)$ for each $y \in \mathbf{R}^k$. It follows from (H4') that $d\mathbf{P} \times dt$ - a.e., for each $n, i \geq 1$ and $y \in \mathbf{R}^k$,

$$\langle y, \tilde{g}^{n,i}(t, y) \rangle \leq u(t)\rho(|y|^2) + |y| |\tilde{g}^{n+i}(t, y_t^n) - \tilde{g}^n(t, y_t^n)|.$$

Moreover, (H3') and (16) indicate that $d\mathbf{P} \times dt$ - a.e., for each $n, i \geq 1$ and $t \in [0, T]$

$$|\tilde{g}^{n+i}(t, y_t^n) - \tilde{g}^n(t, y_t^n)| \leq \psi_{r+1}(t) \mathbf{1}_{\psi_{r+1}(t) > ne^{-t}}.$$

Hence, for each $n, i \geq 1$, $\tilde{g}^{n,i}(t, y)$ satisfies assumption (A) with $\mu(\cdot) = u(\cdot), \kappa(\cdot) = \rho(\cdot), \lambda(\cdot) \equiv 0$ and $f_t = \psi_{r+1}(t) \mathbf{1}_{\psi_{r+1}(t) > ne^{-t}}$. Noticing (16) and Remark 5, by Proposition 4 with $u = 0$ we can get that there exists a constant $C' > 0$ such that for each $t \in [0, T]$,

$$\begin{aligned} & \mathbf{E} \left[|\hat{y}_{t,T}^{n,i,*}|^2 \right] + \mathbf{E} \left[\int_t^T |\hat{z}_s^{n,i}|^2 ds \right] \\ & \leq C' \int_t^T u(s) \rho(\mathbf{E} [|\hat{y}_s^{n,i}|^2]) ds + 2rC' \mathbf{E} \left[\int_t^T \psi_{r+1}(s) \mathbf{1}_{\psi_{r+1}(s) > ne^{-s}} ds \right]. \end{aligned} \quad (17)$$

And in view of (H3'), Lebesgue's dominated convergence theorem yields that, as $n \rightarrow \infty$,

$$\mathbf{E} \left[\int_t^T \psi_{r+1}(s) \mathbf{1}_{\psi_{r+1}(s) > ne^{-s}} ds \right] \rightarrow 0.$$

Then, by taking the supremum with respect to i and then taking the upper limit with respect to n in both sides of (17), in view of Fatou's lemma, the monotonicity and continuity of $\rho(\cdot)$ together with Lemma 3, we know that $\{(y^n, z^n)\}_{n=1}^\infty$ is a Cauchy sequence in the space $\mathcal{S}^2 \times \mathcal{M}^2$. Finally, passing to the limit in both sides of BSDE $(\xi, T, \tilde{g}^n + dV)$ under ucp yields a desired solution of BSDE (9) in $\mathcal{S}^2 \times \mathcal{M}^2$.

Third Step: In this step we will eliminate the additional condition (15), i.e. we will prove that for each $\xi \in L^2(\mathcal{F}_T; \mathbf{R}^k)$ and $V_\cdot \in \mathcal{V}^2$, BSDE (9) admits a solution in $\mathcal{S}^2 \times \mathcal{M}^2$ under assumptions (H1')–(H4').

For each $\gamma \in \mathbf{R}^+$ and $x = (x^1, \dots, x^k) \in \mathbf{R}^k$, we define $\pi_\gamma(x) := (\pi_\gamma^1(x), \dots, \pi_\gamma^k(x))$ with $\pi_\gamma^i(x) := \gamma x^i / (\gamma \vee |x^i|)$ for each $i = 1, \dots, k$. Assume that $V_\cdot = V_\cdot^+ - V_\cdot^-$ with each component of $V_\cdot^+$ and $V_\cdot^-$ being an increasing and $(\mathcal{F}_t)$ -adapted continuous process. For each $n \geq 1$, let

$$\xi^n := \pi_n(\xi), \quad V_t^n := \pi_n(V_t^+) - \pi_n(V_t^-), \quad \hat{g}^n(t, y) := g(t, y) - g(t, 0) + \pi_{ne^{-t}}(g(t, 0)).$$

Clearly, ξ^n , V_t^n and $\hat{g}^n$ satisfy the assumptions in the second step, and by Lebesgue's dominated convergence theorem it is not hard to verify that as $n \rightarrow \infty$,

$$\begin{aligned} \mathbf{E} [|\xi^n - \xi|^2] &\rightarrow 0, \quad \mathbf{E} [|V^n - V|_T^2] \rightarrow 0, \\ \mathbf{E} \left[\left(\int_0^T |\pi_{ne^{-t}}(g(t, 0)) - g(t, 0)| dt \right)^2 \right] &\rightarrow 0. \end{aligned} \quad (18)$$

Now, for each $n \geq 1$, let $(y_t^n, z_t^n)_{t \in [0, T]}$ be a solution of BSDE $(\xi^n, T, \hat{g}^n + dV^n)$. For each $n, m \geq 1$, set $\hat{\xi}^{n,m} := \xi^n - \xi^m$, $\hat{V}^{n,m} := V^n - V^m$, $\hat{y}^{n,m} := y^n - y^m$ and $\hat{z}^{n,m} := z^n - z^m$, then

$$\hat{y}_t^{n,m} = \hat{\xi}^{n,m} + \int_t^T \hat{g}^{n,m}(s, \hat{y}_s^{n,m}) ds + \int_t^T d\hat{V}_s^{n,m} - \int_t^T \hat{z}_s^{n,m} dB_s, \quad t \in [0, T],$$

where $\hat{g}^{n,m}(t, y) := \hat{g}^n(t, y + y_t^m) - \hat{g}^m(t, y_t^m)$ for each $y \in \mathbf{R}^k$. It follows from assumption (H4') that $d\mathbf{P} \times dt$ - a.e., for each $n, m \geq 1$ and $y \in \mathbf{R}^k$,

$$\begin{aligned} \langle y, \hat{g}^{n,m}(t, y) \rangle &= \langle y, \hat{g}^n(t, y + y_t^m) - \hat{g}^m(t, y_t^m) \rangle + \langle y, \hat{g}^n(t, y_t^m) - \hat{g}^m(t, y_t^m) \rangle \\ &\leq u(t)\rho(|y|^2) + |y| |\pi_{ne^{-t}}(g(t, 0)) - \pi_{me^{-t}}(g(t, 0))|, \end{aligned}$$

which means that assumption (A) is satisfied by $\hat{g}^{n,m}(t, y)$ with $\mu(\cdot) = u(\cdot)$, $\kappa(\cdot) = \rho(\cdot)$, $\lambda(\cdot) \equiv 0$ and $f_t = |\pi_{ne^{-t}}(g(t, 0)) - \pi_{me^{-t}}(g(t, 0))|$. Then Proposition 4 with $u = 0$ yields the existence of a constant $\hat{C} > 0$ such that for each $t \in [0, T]$ and $n, m \geq 1$,

$$\begin{aligned} \mathbf{E} [|\hat{y}_{t,T}^{n,m,*}|^2] + \mathbf{E} \left[\int_t^T |\hat{z}_s^{n,m}|^2 ds \right] &\leq \hat{C} \mathbf{E} [|\hat{\xi}^{n,m}|^2] + \hat{C} \mathbf{E} [|\hat{V}_T^{n,m}|^2] \\ &\quad + \hat{C} \int_t^T u(s) \rho \left(\mathbf{E} [|\hat{y}_{s,T}^{n,m,*}|^2] \right) ds \\ &\quad + \hat{C} \mathbf{E} \left[\left(\int_t^T |\pi_{ne^{-t}}(g(t, 0)) - \pi_{me^{-t}}(g(t, 0))| ds \right)^2 \right]. \end{aligned} \quad (19)$$

Furthermore, it follows from Gronwall's inequality (Lemma 2) with $\rho(x) \leq Ax + A$ for each $x \in \mathbf{R}^+$ and the definition of π that

$$\begin{aligned} \mathbf{E} [|\hat{y}_{t,T}^{n,m,*}|^2] + \mathbf{E} \left[\int_t^T |\hat{z}_s^{n,m}|^2 ds \right] &\leq \hat{C} e^{A\hat{C} \int_0^T u(s) ds} \left(A \int_0^T u(s) ds + 4\mathbf{E} [|\xi|^2] + 4\mathbf{E} [|V|_T^2] + 4\mathbf{E} \left[\left(\int_0^T |g(t, 0)|^2 dt \right) \right] \right) \\ &< \infty. \end{aligned}$$

Then taking the upper limit with respect to n and m in both sides of (19) and in view of (18), Fatou's lemma, the monotonicity and continuity of $\rho(\cdot)$ and Bihari's inequality

(Lemma 3), we can obtain that $\{(y^n, z^n)\}_{n=1}^\infty$ is a Cauchy sequence in $\mathcal{S}^2 \times \mathcal{M}^2$. We denote the limit of this sequence in $\mathcal{S}^2 \times \mathcal{M}^2$ by $(y_t, z_t)_{t \in [0, T]}$. Passing to the limit in ucp for BSDE $(\xi^n, T, \hat{g}^n + dV^n)$ yields that $(y_t, z_t)_{t \in [0, T]}$ solves BSDE (9). The proof of Proposition 10 is then completed. $\square$

With Proposition 10 in hand we can prove our main result of this section – Theorem 6.

Proof of the existence part of Theorem 6: Assume now that $\xi \in L^2(\mathcal{F}_T; \mathbf{R}^k)$, $V \in \mathcal{V}^2$ and that (H1)–(H5) hold true for g . We use Picard's iteration method. Let us set as usual $(y^0, z^0) := (0, 0)$ and, by virtue of Proposition 10, define recursively, for each $n \geq 1$,

$$y_t^n = \xi + \int_t^T g(s, y_s^n, z_s^{n-1}) ds + \int_t^T dV_s - \int_t^T z_s^n dB_s, \quad t \in [0, T]. \quad (20)$$

In fact, it is clear that for each $n \geq 1$ and $y \in \mathbf{R}^k$, $g(t, y, z_t^{n-1})$ satisfies (H2') and (H4'). Moreover, for each $n \geq 1$, it follows from (H1), (H5) and Hölder's inequality that

$$\begin{aligned} \mathbf{E} \left[\left(\int_0^T |g(s, 0, z_s^{n-1})| ds \right)^2 \right] &\leq 2\mathbf{E} \left[\left(\int_0^T |g(s, 0, 0)| ds \right)^2 \right] \\ &\quad + 2 \int_0^T v^2(s) ds \cdot \mathbf{E} \left[\int_0^T |z_s^{n-1}|^2 ds \right] < \infty, \end{aligned}$$

which means that $g(t, 0, z_t^{n-1})$ satisfies (H1'). By adding and subtracting terms $g(t, y, 0)$ and $g(t, 0, 0)$ and in view of (H5), we can derive that $d\mathbf{P} \times dt$ – a.e., for each $n \geq 1$ and $r \in \mathbf{R}^+$,

$$\sup_{|y| \leq r} |g(t, y, z_t^{n-1}) - g(t, 0, z_t^{n-1})| \leq 2v(t)|z_t^{n-1}| + \sup_{|y| \leq r} |g(t, y, 0) - g(t, 0, 0)|.$$

Thus, (H3) and Hölder's inequality indicate that for each $n \geq 1$, (H3') holds for $g(t, y, z_t^{n-1})$. Therefore, it follows from Proposition 10 that BSDE (20) admits a solution $(y_t^n, z_t^n)_{t \in [0, T]}$ in $\mathcal{S}^2 \times \mathcal{M}^2$ for each $n \geq 1$. Next we show that this solution sequence is Cauchy in $\mathcal{S}^2 \times \mathcal{M}^2$.

For each $n, i \geq 1$, set $\hat{y}_t^{n,i} := y_t^{n+i} - y_t^n$, $\hat{z}_t^{n,i} := z_t^{n+i} - z_t^n$. Then $(\hat{y}_t^{n,i}, \hat{z}_t^{n,i})_{t \in [0, T]}$ is a solution of the BSDE:

$$\hat{y}_t^{n,i} = \int_t^T \hat{g}^{n,i}(s, \hat{y}_s^{n,i}) ds - \int_t^T \hat{z}_s^{n,i} dB_s, \quad t \in [0, T],$$

where $\hat{g}^{n,i}(t, y) := g(t, y + y_t^n, z_t^{n+i-1}) - g(t, y_t^n, z_t^{n-1})$ for each $y \in \mathbf{R}^k$. It follows from assumptions (H4) and (H5) that $d\mathbf{P} \times dt$ – a.e., for each $n, i \geq 1$ and $y \in \mathbf{R}^k$,

$$\begin{aligned} \langle y, \hat{g}^{n,i}(t, y) \rangle &= \langle y, g(t, y + y_t^n, z_t^{n+i-1}) - g(t, y_t^n, z_t^{n+i-1}) \rangle \\ &\quad + \langle y, g(t, y_t^n, z_t^{n+i-1}) - g(t, y_t^n, z_t^{n-1}) \rangle \\ &\leq u(t)\rho(|y|^2) + v(t)|y||\hat{z}_t^{n-1,i}|, \end{aligned}$$

which guarantees that assumption (A) is satisfied by $\hat{g}^{n,i}$ with $\mu(\cdot) = u(\cdot)$, $\kappa(\cdot) = \rho(\cdot)$, $\lambda(\cdot) \equiv 0$, $f_t = v(t)|\hat{z}_t^{n-1,i}|$. It then follows from Proposition 4 with $u = 0$ and Hölder's

inequality that there exists a constant $C > 0$ such that for each $t \in [0, T]$ and $n, i \geq 1$,

$$\begin{aligned} & \mathbf{E} \left[|\hat{y}_{t,T}^{n,i,*}|^2 \right] + \mathbf{E} \left[\int_t^T |\hat{z}_s^{n,i}|^2 ds \right] \\ & \leq C \left\{ \int_t^T u(s) \rho \left(\mathbf{E} \left[|\hat{y}_{s,T}^{n,i,*}|^2 \right] \right) ds + \int_t^T v^2(s) ds \cdot \mathbf{E} \left[\int_t^T |\hat{z}_s^{n-1,i}|^2 ds \right] \right\}. \end{aligned} \quad (21)$$

In the sequel, in view of the fact that $\int_0^T (u(s) + v^2(s)) ds < \infty$, we can subdivide the interval $[0, T]$, no matter $T < \infty$ or $T = \infty$, into some small intervals like $[T_j, T_{j+1}]$ ($j = 0, 1, \dots, N-1$) such that $0 = T_0 < T_1 < \dots < T_N = T$ with

$$\begin{aligned} T_j &:= \min \left\{ t \in [0, T] : \int_t^{T_{j+1}} u(s) ds \leq \frac{\ln 2}{AC}, \int_t^{T_{j+1}} v^2(s) ds \leq \frac{1}{8C} \right\}, \\ & j = 1, 2, \dots, N-1. \end{aligned}$$

Note that T_{N-1} depends only on A and C and the partition of the bounded interval $[0, T_{N-1}]$ is trivial, the number of small intervals $[T_j, T_{j+1}]$ is finite. Next we will show that for each $j = 0, 1, \dots, N-1$, $\{(y_t^n, z_t^n)_{t \in [T_j, T_{j+1}]}\}_{n=1}^\infty$ is a Cauchy sequence in $\mathcal{S}^2 \times \mathcal{M}^2$. Let T_{j+1} replace the terminal time T in (21). From the fact that $\rho(x) \leq Ax + A$ for each $x \in \mathbf{R}^+$ together with Gronwall's inequality (Lemma 2) we can deduce that for each $j = 0, 1, \dots, N-1$, $t \in [T_j, T_{j+1}]$ and $n, i \geq 1$,

$$\begin{aligned} & \mathbf{E} \left[|\hat{y}_{t,T_{j+1}}^{n,i,*}|^2 \right] + \mathbf{E} \left[\int_t^{T_{j+1}} |\hat{z}_s^{n,i}|^2 ds \right] \\ & \leq e^{AC \int_t^{T_{j+1}} u(s) ds} \left\{ AC \int_t^{T_{j+1}} u(s) ds + C \int_t^{T_{j+1}} v^2(s) ds \mathbf{E} \left[\int_t^{T_{j+1}} |\hat{z}_s^{n-1,i}|^2 ds \right] \right\} \\ & \leq 2AC \int_0^T u(s) ds + \frac{1}{4} \mathbf{E} \left[\int_t^{T_{j+1}} |\hat{z}_s^{n-1,i}|^2 ds \right]. \end{aligned} \quad (22)$$

Therefore, by picking $n = 1$ and $i = m-1$ in the previous inequality we can obtain that for each $m \geq 2$ and $t \in [T_j, T_{j+1}]$ ($j = 0, 1, \dots, N-1$),

$$\begin{aligned} \mathbf{E} \left[\int_t^{T_{j+1}} |z_s^m - z_s^1|^2 ds \right] & \leq 2AC \int_0^T u(s) ds + \frac{1}{4} \mathbf{E} \left[\int_t^{T_{j+1}} |z_s^{m-1}|^2 ds \right] \\ & \leq 2AC \int_0^T u(s) ds + \frac{1}{2} \mathbf{E} \left[\int_0^T |z_s^1|^2 ds \right] \\ & \quad + \frac{1}{2} \mathbf{E} \left[\int_t^{T_{j+1}} |z_s^{m-1} - z_s^1|^2 ds \right], \end{aligned}$$

from which we can calculate by induction that

$$\sup_{m \geq 1} \mathbf{E} \left[\int_t^{T_{j+1}} |z_s^m - z_s^1|^2 ds \right] \leq 4AC \int_0^T u(s) ds + \mathbf{E} \left[\int_0^T |z_s^1|^2 ds \right].$$

Combining the previous inequality with (22) yields that for each $n \geq 1$ and $t \in [T_j, T_{j+1}]$ ($j = 0, 1, \dots, N-1$),

$$\sup_{i \geq 1} \left(\mathbf{E} \left[|\hat{y}_{t, T_{j+1}}^{n, i, *}|^2 \right] + \mathbf{E} \left[\int_t^{T_{j+1}} |\hat{z}_s^{n, i}|^2 ds \right] \right) \leq 4AC \int_0^T u(s) ds + \frac{1}{2} \mathbf{E} \left[\int_0^T |z_s^1|^2 ds \right] < \infty.$$

Thus, taking the supremum with respect to i and then taking the upper limit with respect to n in both sides of (21) with T replaced with T_{j+1} , in view of Fatou's Lemma, the monotonicity and continuity of $\rho(\cdot)$, Bihari's inequality (Lemma 3) and the fact that $\int_t^{T_{j+1}} v^2(s) ds \leq 1/8C$ for each $t \in [T_j, T_{j+1}]$, we know that $\{(y_t^n, z_t^n)\}_{n=1}^\infty$ is a Cauchy sequence in $\mathcal{S}^2(T_j, T_{j+1}; \mathbf{R}^k) \times \mathcal{M}^2(T_j, T_{j+1}; \mathbf{R}^{k \times d})$. Denote the limit process by $(y_t, z_t)_{t \in [T_j, T_{j+1}]}$. After taking the limit under ucp in BSDE (20) we see that there is a solution (y, z) of BSDE (1) on each interval $[T_j, T_{j+1}]$ ($j = 0, 1, \dots, N-1$). The proof is completed. $\square$

Remark 11: From the above proof procedure we can observe that in the one-dimensional case Theorem 6 remains valid when $(y_t)_{t \in [0, T]}$ and $(V_t)_{t \in [0, T]}$ are assumed to be only RCLL processes.

3.3. Stochastic time interval case

This subsection asserts an stochastic interval version of Theorem 6, namely that if the terminal time T is replaced by a stopping time valued in $[0, T]$, the existence and uniqueness result remains valid. As said in the introduction, this result is not trivial, and it will be used in the following Sections 4 and 5. The proof is inspired by Corollary 2.1 in Yao [26].

Theorem 12: Assume that $0 \leq T \leq \infty$, $(V_t)_{t \in [0, T]} \in \mathcal{V}^2$, τ is a $(\mathcal{F}_t)$ -stopping time valued in $[0, T]$, $\xi \in L^2(\mathcal{F}_\tau, \mathbf{R}^k)$, and the generator g satisfies (H1)–(H5). Define $g_\tau(\omega, t, y, z) := \mathbf{1}_{t \leq \tau} g(\omega, t, y, z)$ for each $(\omega, t, y, z) \in \Omega \times [0, T] \times \mathbf{R}^k \times \mathbf{R}^{k \times d}$, and $\bar{V}_t := V_t \mathbf{1}_{t \leq \tau} + V_\tau \mathbf{1}_{t > \tau}$ for each $t \in [0, T]$. Then the following three assertions hold:

- (i) BSDE $(\xi, T, g_\tau + d\bar{V})$ admits a unique solution $(\bar{y}_t, \bar{z}_t)_{t \in [0, T]}$ in $\mathcal{S}^2 \times \mathcal{M}^2$, $\bar{y}_t \mathbf{1}_{t \geq \tau} = \xi$ and $\bar{z}_t \mathbf{1}_{t \geq \tau} = 0$, $d\mathbf{P} \times dt$ – a.e.. In particular, $(y_t, z_t)_{t \in [0, T]} := (\bar{y}_t, \bar{z}_t)_{t \in [0, T]}$ solves the following BSDE $(\xi, \tau, g + dV)$ in $\mathcal{S}^2 \times \mathcal{M}^2$:

$$y_t = \xi + \int_t^\tau g(s, y_s, z_s) ds + \int_t^\tau dV_s - \int_t^\tau z_s dB_s, \quad t \in [0, \tau];$$

- (ii) Let $(y_t, z_t)_{t \in [0, T]} \in \mathcal{S}^2 \times \mathcal{M}^2$ satisfy BSDE $(\xi, \tau, g + dV)$. Then $(\bar{y}_t, \bar{z}_t)_{t \in [0, T]}$ is a solution of BSDE $(\xi, T, g + d\bar{V})$ in $\mathcal{S}^2 \times \mathcal{M}^2$, where $\bar{y}_t := y_t \mathbf{1}_{t \leq \tau} + \xi \mathbf{1}_{t > \tau}$ and $\bar{z}_t := z_t \mathbf{1}_{t \leq \tau}$ for each $t \in [0, T]$;
- (iii) BSDE $(\xi, \tau, g + dV)$ admits a unique solution $(y_t, z_t)_{t \in [0, T]}$ in $\mathcal{S}^2 \times \mathcal{M}^2$, i.e. if $(y'_t, z'_t)_{t \in [0, T]} \in \mathcal{S}^2 \times \mathcal{M}^2$ also solves BSDE $(\xi, \tau, g + dV)$, then $y_t \mathbf{1}_{t \leq \tau} = y'_t \mathbf{1}_{t \leq \tau}$ and $z_t \mathbf{1}_{t \leq \tau} = z'_t \mathbf{1}_{t \leq \tau}$, $d\mathbf{P} \times dt$ – a.e..

Proof: Let assumptions hold. It is clear that g_τ is also a generator satisfying (H1)–(H5) and $\bar{V} \in \mathcal{V}^2$. Then an immediate consequence of Theorem 6 is that BSDE $(\xi, T, g_\tau + d\bar{V})$, which can be written as

$$\bar{y}_t = \xi + \int_t^T \mathbf{1}_{s \leq \tau} g(s, \bar{y}_s, \bar{z}_s) ds + \int_t^T d\bar{V}_s - \int_t^T \bar{z}_s dB_s, \quad t \in [0, T], \quad (23)$$

admits a unique solution $(\bar{y}_t, \bar{z}_t)_{t \in [0, T]}$ in $\mathcal{S}^2 \times \mathcal{M}^2$. Then we get that

$$\begin{aligned} \bar{y}_\tau &= \xi + \int_\tau^T \mathbf{1}_{s \leq \tau} g(s, \bar{y}_s, \bar{z}_s) ds + \int_\tau^T d\bar{V}_s - \int_\tau^T \bar{z}_s dB_s \\ &= \xi - \int_\tau^T \bar{z}_s dB_s. \end{aligned}$$

Taking conditional expectation $\mathbf{E}[\cdot | \mathcal{F}_\tau]$ in the above identity yields that $\bar{y}_\tau = \mathbf{E}[\xi | \mathcal{F}_\tau] = \xi$, and then

$$\int_\tau^T \bar{z}_s dB_s = \int_0^T \bar{z}_s \mathbf{1}_{s > \tau} dB_s = 0,$$

which implies that $\mathbf{E}[\int_0^T |\bar{z}_s \mathbf{1}_{s > \tau}|^2 ds] = 0$. Hence, $\bar{z}_t \mathbf{1}_{t > \tau} = 0$, $d\mathbf{P} \times dt$ - a.e., and $\bar{y}_t \mathbf{1}_{t > \tau} = \xi$ by (23). Thus, noticing that $\bar{y}_\tau = \xi$, we have

$$\bar{y}_t = \xi + \int_t^\tau \mathbf{1}_{s \leq \tau} g(s, \bar{y}_s, \bar{z}_s) ds + \int_t^\tau d\bar{V}_s - \int_t^\tau \bar{z}_s dB_s, \quad t \in [0, \tau],$$

from which it follows immediately that $(y_t, z_t)_{t \in [0, T]} := (\bar{y}_t, \bar{z}_t)_{t \in [0, T]}$ solves BSDE $(\xi, \tau, g + dV)$. Assertion (i) is then proved. Furthermore, (ii) can be directly verified so that we omit its proof. Finally, in view of the existence and uniqueness result for solutions of BSDE $(\xi, T, g_\tau + d\bar{V})$, (iii) is a direct consequence of assertions (i) and (ii). $\square$

4. Comparison theorem for solutions of one-dimensional BSDEs

In this section, we only consider the one-dimensional BSDE, i.e. $k = 1$ in BSDE (1), and prove a general comparison theorem for solutions, which extends the corresponding result in Fan and Jiang [9] to the general time interval BSDE.

Theorem 13: Assume that $0 \leq T \leq \infty$, σ and τ are two $(\mathcal{F}_t)$ -stopping times with $0 \leq \sigma \leq \tau \leq T$, ξ and $\xi' \in L^2(\mathcal{F}_\tau; \mathbf{R})$, V and $V' \in \mathcal{V}^2(0, T; \mathbf{R})$, g and g' are two generators of BSDEs, and (y, z) (resp. (y', z')) is a solution of BSDE $(\xi, \tau, g + dV)$ (resp. $(\xi', \tau, g' + dV')$). If

- (i) $\xi \leq \xi'$;
- (ii) g satisfies (H4)–(H5) and $d\mathbf{P} \times dt$ - a.e., $g(t, y'_t, z'_t) \leq g'(t, y'_t, z'_t)$;
- (iii) $(V_t - V'_t)_{t \in [0, T]}$ is decreasing on the interval $[\sigma, \tau]$,

then for each $(\mathcal{F}_t)$ -stopping time δ with $\delta \in [\sigma, \tau]$, we have $y_\delta \leq y'_\delta$. Particularly, $y_\sigma \leq y'_\sigma$.

Proof: The main arguments are enlightened by Fan and Jiang [9] and Cao and Yan [4]. Here we just give the sketch of this proof highlighting the difference. Let the assumptions

in Theorem 13 hold and set $\hat{\xi} := \xi - \xi'$, $\hat{y} := y - y'$, $\hat{z} := z - z'$, $\hat{V} := V - V'$. Then we have

$$\hat{y}_t = \hat{\xi} + \int_t^\tau (g(s, y_s, z_s) - g'(s, y'_s, z'_s)) ds + \int_t^\tau d\hat{V}_s - \int_t^\tau \langle \hat{z}_s, dB_s \rangle, \quad t \in [0, \tau].$$

It follows from Lemma 2.2 in Cao and Yan [4] that for each $t \in [0, \tau]$,

$$\begin{aligned} (\hat{y}_{t \wedge \tau \vee \sigma}^+)^2 + \int_{t \wedge \tau \vee \sigma}^\tau \mathbf{1}_{\hat{y}_s > 0} |\hat{z}_s|^2 ds &= (\hat{\xi}^+)^2 + 2 \int_{t \wedge \tau \vee \sigma}^\tau \hat{y}_s^+ (g(s, y_s, z_s) - g'(s, y'_s, z'_s)) ds \\ &\quad + 2 \int_{t \wedge \tau \vee \sigma}^\tau \hat{y}_s^+ d\hat{V}_s - 2 \int_{t \wedge \tau \vee \sigma}^\tau \hat{y}_s^+ \langle \hat{z}_s, dB_s \rangle. \end{aligned} \quad (24)$$

By adding and subtracting terms $g(s, y'_s, z'_s)$ and $g(s, y_s, z_s)$, and in view of the fact that $g(s, y'_s, z'_s) - g'(s, y'_s, z'_s)$ is non-positive, and (H4)–(H5), we can deduce that $d\mathbf{P} \times dt$ - a.e.,

$$2\hat{y}_s^+ (g(s, y_s, z_s) - g'(s, y'_s, z'_s)) \leq 2u(s)\rho((\hat{y}_s^+)^2) + v^2(s)(\hat{y}_s^+)^2 + \mathbf{1}_{\hat{y}_s > 0} |\hat{z}_s|^2. \quad (25)$$

Plugging the previous inequality into (24) and then taking expectation, in view of the facts that $\hat{\xi} \leq \xi'$, $(\hat{V}_t)_{t \in [0, T]}$ is decreasing on $[\sigma, \tau]$ and $(\int_0^t \hat{y}_s^+ \hat{z}_s dB_s)_{t \in [0, T]}$ is a $(\mathcal{F}_t)$ -martingale, we can deduce by Fubini's theorem and Jensen's inequality that for each $t \in [0, T]$,

$$\mathbf{E}[(\hat{y}_{t \wedge \tau \vee \sigma}^+)^2] \leq \mathbf{E}\left[\int_{t \wedge \tau \vee \sigma}^\tau \bar{u}(s)\bar{\rho}((\hat{y}_s^+)^2) ds\right] \leq \int_t^T \bar{u}(s)\bar{\rho}(\mathbf{E}[(\hat{y}_{s \wedge \tau \vee \sigma}^+)^2]) ds,$$

where $\bar{u}(s) := 2u(s) + v^2(s)$, $\bar{\rho}(x) := \rho(x) + x$ for each $x \in \mathbf{R}^+$. Since $\bar{\rho} \in \mathbb{S}$, Bihari's inequality (Lemma 3) yields that for each $t \in [0, T]$, $\mathbf{E}[(\hat{y}_{t \wedge \tau \vee \sigma}^+)^2] = 0$ and then $\hat{y}_{t \wedge \tau \vee \sigma}^+ = 0$. Furthermore, in view of the continuity of $\hat{y}^+$, we know that $\hat{y}_{t \wedge \tau \vee \sigma}^+ = 0$, $t \in [0, T]$, and then $\hat{y}_\delta^+ = 0$ for each $(\mathcal{F}_t)$ -stopping time $\delta \in [\sigma, \tau]$, i.e. $y_\delta \leq y'_\delta$. Particularly, $y_\sigma \leq y'_\sigma$. $\square$

Remark 14: Theorem 13 remains valid if the assumption (ii) is replaced by

(ii') g' satisfies (H4)–(H5) and $d\mathbf{P} \times dt$ - a.e., $g(t, y_t, z_t) \leq g'(t, y_t, z_t)$.

Indeed, with (ii') in hand, (25) holds still true, then the same argument as above yields the result.

5. Nonlinear Doob-Meyer's decomposition for general continuous g -supermartingales

This section aims at adopting Theorems 6 and 13 to prove a nonlinear Doob-Meyer's decomposition theorem for general continuous g -supermartingales, in which the conditions required by g is weaker than those in Peng [21].

Now we assume that $0 \leq T \leq \infty$, τ is a $(\mathcal{F}_t)$ -stopping time valued in $[0, T]$, the generator $g : \Omega \times [0, T] \times \mathbf{R} \times \mathbf{R}^d \mapsto \mathbf{R}$ is $(\mathcal{F}_t)$ -progressively measurable for each $(y, z) \in \mathbf{R} \times \mathbf{R}^d$ and satisfies (H1)–(H5). We first give the notion of g -supersolution in the case of $0 \leq T \leq \infty$.

Definition 15: Assume that there exists an increasing continuous process $(V_t)_{t \in [0, T]} \in \mathcal{V}^2(0, T; \mathbf{R})$ such that $(y_t, z_t)_{t \in [0, T]}$ is a solution of BSDE $(y_\tau, \tau, g + dV)$, where τ is a

$(\mathcal{F}_t)$ -stopping time valued in $[0, T]$. Then $y_\cdot$ is called a g -supersolution on $[0, \tau]$ (with terminal condition y_τ). Furthermore, $y_\cdot$ is called a g -solution on $[0, \tau]$ if $V \equiv 0$ on $[0, \tau]$.

By Theorem 6 we know that a g -supersolution $y_\cdot$ on $[0, T]$ with terminal condition $y_T \in L^2(\mathcal{F}_T; \mathbf{R})$ is uniquely determined if $(V_t)_{t \in [0, T]}$ are given. Moreover, similar to the proof of Proposition 1.6 in Peng [21] we can also check that the pair of processes $(z_t, V_t)_{t \in [0, T]}$ is uniquely determined by the g -supersolution $y_\cdot$ on $[0, T]$ with terminal condition $y_T \in L^2(\mathcal{F}_T; \mathbf{R})$. Thus, we can define

Definition 16: Let $y_\cdot$ be a g -supersolution on $[0, T]$ with terminal condition $y_T \in L^2(\mathcal{F}_T; \mathbf{R})$ and let $(y_t, z_t, V_t)_{t \in [0, T]}$ be the related unique triple in the sense of BSDE $(y_T, T, g + dV)$. Then we call $(z_\cdot, V_\cdot)$ the (unique) decomposition of $y_\cdot$ on $[0, T]$.

In the sequel, we introduce the notions of g -martingales and g -supermartingales (g -submartingales), where $0 \leq T \leq \infty$.

Definition 17: A process $Y_\cdot \in \mathcal{S}^2(0, T; \mathbf{R})$ is called a continuous g -martingale on $[0, T]$ if it is a g -solution on $[0, T]$ with terminal condition Y_T . A process $Y_\cdot \in \mathcal{S}^2(0, T; \mathbf{R})$ is called a continuous g -supermartingale (resp. g -submartingale) on $[0, T]$ if for each $(\mathcal{F}_t)$ -stopping time τ with $0 \leq \tau \leq T$, g -solution $y_\cdot$ on $[0, \tau]$ with terminal condition $y_\tau = Y_\tau$ satisfies $y_\sigma \leq Y_\sigma$ (resp. $y_\sigma \geq Y_\sigma$) for all $(\mathcal{F}_t)$ -stopping time σ with $0 \leq \sigma \leq \tau$.

By Theorem 13 we can easily check that a g -supersolution on $[0, T]$ is also a g -supermartingale. And the inverse problem is solved by the following Theorem 18, i.e. a continuous g -supermartingale on $[0, T]$ is also a g -supersolution on $[0, T]$, which is indeed a nonlinear version of Doob-Meyer's decomposition theorem. To state it we need the following additional assumption:

(H6) $dP \times dt$ - a.e., for each $y \in \mathbf{R}$, $g(t, y, 0) \geq -g_t - u(t)|y|$, where $u(\cdot)$ is defined in Section 3.1, and $(g_t)_{t \in [0, T]}$ is a nonnegative and $(\mathcal{F}_t)$ -progressively measurable process with $E[(\int_0^T g_t dt)^2] < \infty$.

Theorem 18: Assume that (H1)–(H6) hold and $Y_\cdot \in \mathcal{S}^2(0, T; \mathbf{R})$ is a continuous g -supermartingale on $[0, T]$. Then $Y_\cdot$ is a g -supersolution on $[0, T]$ with terminal condition Y_T , i.e. there exists a unique continuous increasing process $(V_t)_{t \in [0, T]} \in \mathcal{V}^2(0, T; \mathbf{R})$ such that $(Y_t)_{t \in [0, T]}$ coincides with $(y_t)_{t \in [0, T]}$, where $(y_t, z_t)_{t \in [0, T]}$ is the unique solution of BSDE $(Y_T, T, g + dV)$.

Proof: Some idea of the proof is enlightened by Peng [21] and Shi et al. [25]. However, some new difficulties arise naturally here since T may be infinite and the generator g can have a general growth in y in our setting. Now, let the assumptions in Theorem 18 hold. By Theorem 6 there exists, for each $n \geq 1$, a unique solution $(y_t^n, z_t^n)_{t \in [0, T]}$ for the following BSDE:

$$y_t^n = Y_T + \int_t^T g(s, y_s^n, z_s^n) ds + \int_t^T dV_s^n - \int_t^T \langle z_s^n, dB_s \rangle, \quad t \in [0, T], \quad (26)$$

where

$$V_t^n := n \int_0^t e^{-s} (Y_s - y_s^n) ds.$$

Indeed, BSDE (26) can be rewritten as the following type, for each $n \geq 1$,

$$y_t^n = Y_T + \int_t^T (g(s, y_s^n, z_s^n) - ne^{-s}y_s^n) ds + \int_t^T d\bar{V}_s^n - \int_t^T \langle z_s^n, dB_s \rangle, \quad t \in [0, T],$$

where $\bar{V}_t^n := n \int_0^t e^{-s} Y_s ds$. It is not hard to verify that the generator $\bar{g}_n(t, y, z) := g(t, y, z) - ne^{-t}y$ fulfills assumptions (H1)–(H5) and $(\bar{V}_t^n)_{t \in [0, T]}$ belongs to $\mathcal{V}^2(0, T; \mathbf{R})$ for each $n \geq 1$. Then Theorem 6 applies. Next we will divide the proof into four steps.

First Step: This step aims at proving the assertion that for each $n \geq 1$ and $t \in [0, T]$, $y_t^n \leq Y_t$.

Here we use an analogue argument to that of Lemma 3.4 in Peng [21]. If the assertion is not true, then there exists a $n \geq 1$ and a $\delta > 0$ such that $\mathbf{P}(\sigma < T) > 0$, where the $(\mathcal{F}_t)$ -stopping time σ is defined as follows, with the convention that $\inf \emptyset = \infty$,

$$\sigma := \inf\{t \geq 0 : y_t^n \geq Y_t + \delta\} \wedge T.$$

Now we define the following $(\mathcal{F}_t)$ -stopping time:

$$\tau := \inf\{t \geq \sigma : y_t^n \leq Y_t\},$$

which satisfies $\sigma \leq \tau \leq T$ since $y_T^n = Y_T$. Then, $\{\sigma < T\} = \{\sigma < \tau\} =: B$ and $\mathbf{P}(B) > 0$. Since $(Y_t - y_t^n)_{t \in [0, T]}$ is continuous, we get that

$$(i) \mathbf{1}_B y_\sigma^n \geq \mathbf{1}_B Y_\sigma + \delta \mathbf{1}_B, \quad \text{and} \quad (ii) y_\tau^n \leq Y_\tau. \quad (27)$$

Next we construct a g -solution y' on $[0, \tau]$ with terminal condition Y_τ as follows:

$$y'_t = Y_\tau + \int_t^\tau g(s, y'_s, z'_s) ds - \int_t^\tau \langle z'_s, dB_s \rangle, \quad t \in [0, \tau]. \quad (28)$$

Note that Y is a continuous g -supermartingale on $[0, T]$, it follows from Definition 17 with (28) that $y'_\sigma \leq Y_\sigma$. Moreover, it follows from (26) that

$$y_t^n = y_\tau^n + \int_t^\tau g(s, y_s^n, z_s^n) ds + \int_t^\tau dV_s^n - \int_t^\tau \langle z_s^n, dB_s \rangle, \quad t \in [0, \tau]. \quad (29)$$

Since $y_t^n \geq Y_t$ holds on $[\sigma, \tau]$, we know that $(V_t^n)_{t \in [0, T]}$ is decreasing on $[\sigma, \tau]$. Then, in view of (ii) in (27), (28) and (29), it follows from Theorem 13 that $y_\sigma^n \leq y'_\sigma$. Hence, we have $y_\sigma^n \leq Y_\sigma$. But this contradicts with (i) in (27). Thus, we have proved that for each $n \geq 1$ and $t \in [0, T]$, $y_t^n \leq Y_t$.

Second Step: In this step we shall prove that the triple $(y_t^n, z_t^n, V_t^n)_{t \in [0, T]}$ is uniformly bounded in n on $\mathcal{S}^2 \times \mathcal{M}^2 \times \mathcal{V}^2$, i.e.

$$\sup_{n \geq 1} \left(\mathbf{E}[|y_T^{n,*}|^2] + \mathbf{E} \left[\int_0^T |z_s^n|^2 ds \right] + \mathbf{E}[|V_T^n|^2] \right) < \infty. \quad (30)$$

It follows from the first step that $V^n \in \mathcal{V}^2(0, T; \mathbf{R})$ is an increasing process for each $n \geq 1$. Then for each $n \geq 1$, y^n is a g -supersolution on $[0, T]$. Furthermore, it follows from Theorem 13 that for each $n \geq 1$ and $t \in [0, T]$, $y_t^n \leq y_t^{n+1}$, and then, in view of the first

step,

$$|y_t^n| \leq |y_t^1| + |Y_t| \leq \sup_{t \in [0, T]} (|y_t^1| + |Y_t|) =: M. \quad (31)$$

Thus, $(y_t^n)_{t \in [0, T]}$ is uniformly bounded in n on $\mathcal{S}^2(0, T; \mathbf{R})$, i.e.

$$\sup_{n \geq 1} \mathbf{E} [|y_T^{n,*}|^2] \leq \mathbf{E} [|M|^2] < \infty. \quad (32)$$

In the sequel, it follows from (26), (H5) and (H6) that for each $n \geq 1$,

$$|V_T^n| \leq |y_0^n| + |Y_T| + \int_0^T (g_s + u(s)|y_s^n|) \, ds + \int_0^T v(s)|z_s^n| \, ds + \left| \int_0^T \langle z_s^n, dB_s \rangle \right|.$$

Then, in view of (32), taking the square and the mathematical expectation in both sides of the previous inequality yields that for each $n \geq 1$,

$$\begin{aligned} \mathbf{E} [|V_T^n|^2] &\leq c_1 \mathbf{E} [|y_T^{n,*}|^2] + c_1 \mathbf{E} [|Y_T|^2] + c_1 \mathbf{E} \left[\left(\int_0^T g_s \, ds \right)^2 \right] + c_1 \mathbf{E} \left[\int_0^T |z_s^n|^2 \, ds \right] \\ &\leq c_2 + c_1 \mathbf{E} \left[\int_0^T |z_s^n|^2 \, ds \right], \end{aligned} \quad (33)$$

where BDG's inequality and Hölder's inequality have been used, c_1 and c_2 are two non-negative constants. On the other hand, (H4) and (H5) imply that $d\mathbf{P} \times dt$ — a.e., for each $t \in [0, T]$, $y \in \mathbf{R}$ and $z \in \mathbf{R}^d$,

$$yg(t, y, z) \leq u(t)\rho(|y|^2) + v(t)|y||z| + |y||g(t, 0, 0)|,$$

which means that assumption (A) is satisfied by g with $\mu(\cdot) = u(\cdot)$, $\kappa(\cdot) = \rho(\cdot)$, $\lambda(\cdot) = v(\cdot)$ and $f_t = |g(t, 0, 0)|$. A similar argument to that of obtaining (5) yields that for each $n \geq 1$,

$$\begin{aligned} \mathbf{E} \left[\int_0^T |z_s^n|^2 \, ds \right] &\leq 2\mathbf{E} [|Y_T|^2] + 4\mathbf{E} \left[\int_0^T u(s)\rho(|y_s^n|^2) \, ds \right] + 4 \int_0^T v^2(s) \, ds \mathbf{E} [|y_T^{n,*}|^2] \\ &\quad + 4\mathbf{E} \left[\int_0^T y_s^n \, dV_s^n \right] + 4\mathbf{E} \left[\int_0^T |y_s^n| |g(s, 0, 0)| \, ds \right]. \end{aligned} \quad (34)$$

Moreover, by virtue of $2ab \leq \alpha a^2 + b^2/\alpha$ for $\alpha > 0$, we have

$$\begin{aligned} 4\mathbf{E} \left[\int_0^T y_s^n \, dV_s^n \right] &\leq 16c_1 \mathbf{E} [|y_T^{n,*}|^2] + \frac{1}{4c_1} \mathbf{E} [|V_T^n|^2], \\ 4\mathbf{E} \left[\int_0^T |y_s^n| |g(s, 0, 0)| \, ds \right] &\leq 2\mathbf{E} [|y_T^{n,*}|^2] + 2\mathbf{E} \left[\left(\int_0^T |g(s, 0, 0)| \, ds \right)^2 \right]. \end{aligned}$$

Thus, since $\rho(x) \leq Ax + A$ for each $x \in \mathbf{R}^+$, by plugging the previous two inequalities into (34) and using (32) we can conclude that there exists a nonnegative constant c_3 such that for each $n \geq 1$,

$$\mathbf{E} \left[\int_0^T |z_s^n|^2 ds \right] \leq c_3 + \frac{1}{4c_1} \mathbf{E} [|V_T^n|^2].$$

Combining the previous inequality with (33) yields that $\mathbf{E} [|V_T^n|^2]$ and $\mathbf{E} [\int_0^T |z_s^n|^2 ds]$ are also uniformly bounded in n , i.e.

$$\sup_{n \geq 1} \mathbf{E} \left[\int_0^T |z_s^n|^2 ds \right] + \sup_{n \geq 1} \mathbf{E} [|V_T^n|^2] < \infty.$$

Third Step: In this step we will prove that the limit of the process sequence $\{y_t^n\}_{n=1}^\infty$ is a RCLL process. Here, we emphasize that those methods and techniques used in Shi et al. [25] and Peng [21] can not work since T may be infinite and g can have a general growth in y in our setting.

We will adopt the stopping time tool and a weak convergence technique in a new process space. First, for each $q \geq 1$, we introduce the following $(\mathcal{F}_t)$ -stopping time:

$$\tau_q := \inf \{t \geq 0 : |y_t^1| + |Y_t| \geq q\} \wedge T.$$

It follows from (H3) and (H5) together with (31) that $d\mathbf{P} \times dt - \text{a.e.}$, for each $q \geq 1$ and $n \geq 1$,

$$\begin{aligned} |g(t, y_t^n, z_t^n) \mathbf{1}_{t \leq \tau_q}| &\leq |g(t, y_t^n, 0) - g(t, 0, 0)| \mathbf{1}_{t \leq \tau_q} + |g(t, 0, 0)| + v(t) |z_t^n| \\ &\leq \psi_q(t) + |g(t, 0, 0)| + v(t) |z_t^n|. \end{aligned} \quad (35)$$

Hölder's inequality and (30) result in

$$\sup_{n \geq 1} \mathbf{E} \left[\int_0^T v(t) |z_t^n| dt \right] \leq \left(\int_0^T v^2(t) dt \right)^{1/2} \sup_{n \geq 1} \left(\mathbf{E} \left[\int_0^T |z_t^n|^2 dt \right] \right)^{1/2} < \infty.$$

Moreover, let Q be a measurable subset of $\Omega \times [0, T]$. By Hölder's inequality, (30) and Lebesgue's dominated convergence theorem we know that, as $\delta \rightarrow 0$,

$$\begin{aligned} &\sup_{d\mathbf{P} \times dt(Q) < \delta} \sup_{n \geq 1} \mathbf{E} \left[\int_0^T v(s) |z_s^n| \mathbf{1}_Q ds \right] \\ &\leq \left(\sup_{n \geq 1} \mathbf{E} \left[\int_0^T |z_s^n|^2 ds \right] \right)^{1/2} \cdot \sup_{d\mathbf{P} \times dt(Q) < \delta} \left(\mathbf{E} \left[\int_0^T v^2(s) \mathbf{1}_Q ds \right] \right)^{1/2} \rightarrow 0. \end{aligned}$$

Therefore, the process sequence $\{(v(t) |z_t^n|)_{t \in [0, T]}\}_{n=1}^\infty$ is uniformly integrable on $\Omega \times [0, T]$ with respect to the product measure $d\mathbf{P} \times dt$, then so is $\{(g(t, y_t^n, z_t^n) \mathbf{1}_{t \leq \tau_q})_{t \in [0, T]}\}_{n=1}^\infty$ in view of (35) and the fact that $\mathbf{E} [\int_0^T \psi_q(t) dt] < \infty$ and $\mathbf{E} [\int_0^T |g(t, 0, 0)| dt] < \infty$. Thus, we can claim that $\{(g(t, y_t^n, z_t^n) \mathbf{1}_{t \leq \tau_q})_{t \in [0, T]}\}_{n=1}^\infty$ is weakly sequentially compact in a process space $\mathcal{H}$, which implies that there exists a subsequence of $\{(g(t, y_t^n, z_t^n) \mathbf{1}_{t \leq \tau_q})_{t \in [0, T]}\}_{n=1}^\infty$, still

denoted by itself, converges weakly in $\mathcal{H}(0, T; \mathbf{R})$ to a process $({}^q\eta_t)_{t \in [0, T]} \in \mathcal{H}(0, T; \mathbf{R})$, where $\mathcal{H}(0, T; \mathbf{R})$ (or $\mathcal{H}$) denotes the set of (equivalent classes of) $(\mathcal{F}_t)$ -progressively measurable real valued processes $(X_t)_{t \in [0, T]}$ such that

$$\|X\|_{\mathcal{H}} := \mathbf{E} \left[\int_0^T |X_t| dt \right] < \infty.$$

Then, for any bounded linear functional $F(\cdot)$ defined on $L^1(\mathcal{F}_T; \mathbf{R})$, which is a set of real valued and $\mathcal{F}_T$ -measurable random variables Z such that $\|Z\|_{L^1} := \mathbf{E}[|Z|] < \infty$, there exists a positive constant c such that for any $X \in \mathcal{H}$ and any $(\mathcal{F}_t)$ -stopping time σ with $0 \leq \sigma \leq \tau_q$,

$$F \left(\int_0^\sigma X_t dt \right) \leq c \mathbf{E} \left[\left| \int_0^\sigma X_t dt \right| \right] \leq c \mathbf{E} \left[\int_0^T |X_t| dt \right] = c \|X\|_{\mathcal{H}}.$$

This indicates that $F(\int_0^\sigma \cdot dt)$ is a bounded linear functional on $\mathcal{H}(0, T; \mathbf{R})$. Thus, since the sequence $\{(g(t, y_t^n, z_t^n) \mathbf{1}_{t \leq \tau_q})_{t \in [0, T]}\}_{n=1}^\infty$ converges weakly to $({}^q\eta_t)_{t \in [0, T]}$ in $\mathcal{H}(0, T; \mathbf{R})$, we can assert that for each $q \geq 1$ and $(\mathcal{F}_t)$ -stopping time σ with $0 \leq \sigma \leq \tau_q$, as $n \rightarrow \infty$, $\int_0^\sigma g(t, y_t^n, z_t^n) dt = \int_0^\sigma g(t, y_t^n, z_t^n) \mathbf{1}_{t \leq \tau_q} dt$ converges weakly to $\int_0^\sigma {}^q\eta_t dt$ in $L^1(\mathcal{F}_T; \mathbf{R})$, which is rewritten as follows,

$$\int_0^\sigma g(t, y_t^n, z_t^n) dt \xrightarrow{\text{weak}} \int_0^\sigma {}^q\eta_t dt \quad \text{in } L^1(\mathcal{F}_T; \mathbf{R}). \quad (36)$$

On the other hand, it follows from (30) that there exists a subsequence of $\{z_t^n\}_{n=1}^\infty$, still denoted by itself, that converges weakly in $\mathcal{M}^2(0, T; \mathbf{R}^d)$ to a process $(z_t)_{t \in [0, T]}$. Then, for any bounded linear functional $\bar{F}(\cdot)$ defined on $L^2(\mathcal{F}_T; \mathbf{R})$, there exists a positive constant $\bar{c}$ such that for each $(\bar{z}_t)_{t \in [0, T]} \in \mathcal{M}^2(0, T; \mathbf{R}^d)$ and $(\mathcal{F}_t)$ -stopping time σ with $0 \leq \sigma \leq \tau_q$,

$$\bar{F} \left(\int_0^\sigma \langle \bar{z}_t, dB_t \rangle \right) \leq \bar{c} \left\| \int_0^\sigma \langle \bar{z}_t, dB_t \rangle \right\|_{L^2} \leq \bar{c} \|\bar{z}\|_{\mathcal{M}^2}.$$

This indicates that $\bar{F}(\int_0^\sigma \langle \cdot, dB_t \rangle)$ is a bounded linear functional on $\mathcal{M}^2(0, T; \mathbf{R}^d)$. Accordingly, we can assert that for each $q \geq 1$ and $(\mathcal{F}_t)$ -stopping time σ with $0 \leq \sigma \leq \tau_q$, as $n \rightarrow \infty$,

$$\int_0^\sigma \langle z_t^n, dB_t \rangle \xrightarrow{\text{weak}} \int_0^\sigma \langle z_t, dB_t \rangle \quad \text{in } L^2(\mathcal{F}_T; \mathbf{R}) \text{ and then in } L^1(\mathcal{F}_T; \mathbf{R}). \quad (37)$$

Finally, since for each $t \in [0, T]$ and $n \geq 1$, $y_t^n \leq y_t^{n+1} \leq Y_t$, it follows that there exists a process $(y_t)_{t \in [0, T]}$ satisfying $\mathbf{E}[\sup_{t \in [0, T]} |y_t|^2] < \infty$ such that $d\mathbf{P} \times dt$ - a.e., $y_t^n \rightarrow y_t$ as $n \rightarrow \infty$. Therefore, for each $q \geq 1$ and $(\mathcal{F}_t)$ -stopping time σ with $0 \leq \sigma \leq \tau_q$, y_σ^n converges to y_σ in $L^1(\mathcal{F}_T; \mathbf{R})$.

Next, for each $q \geq 1$, define

$${}^qV_t := y_0 - y_t - \int_0^t {}^q\eta_s ds + \int_0^t \langle z_s, dB_s \rangle, \quad t \in [0, T].$$

Then with the help of (36), (37) and the fact that y_σ^n converges to y_σ in $L^1(\mathcal{F}_T; \mathbf{R})$, we can deduce that for each $q \geq 1$ and $(\mathcal{F}_t)$ -stopping time σ with $0 \leq \sigma \leq \tau_q$, the following random variable sequence

$$V_\sigma^n = y_0^n - y_\sigma^n - \int_0^\sigma g(s, y_s^n, z_s^n) ds + \int_0^\sigma \langle z_s^n, dB_s \rangle,$$

converges weakly to ${}^qV_\sigma$ in $L^1(\mathcal{F}_T; \mathbf{R})$ as $n \rightarrow \infty$. Moreover, for each $n, q \geq 1$ and any two $(\mathcal{F}_t)$ -stopping times σ and τ with $0 \leq \sigma \leq \tau \leq \tau_q$, due to $V_\sigma^n \leq V_\tau^n$ it follows that ${}^qV_\sigma \leq {}^qV_\tau$, and in view of the definition of qV and the facts that $y^n \leq y^{n+1} \leq Y$ and $(y_t^n)_{t \in [0, T]} \in \mathcal{S}^2(0, T; \mathbf{R})$ for each $n \geq 1$, it is not hard to verify that qV is a $(\mathcal{F}_t)$ -optional process with $\mathbf{P}$ — a.s. upper semi-continuous paths. Thus, Lemma A.3 in Bayraktar and Yao [1] yields that for each $q \geq 1$, ${}^qV_{\cdot \wedge \tau_q}$ is an increasing process, and then it has $\mathbf{P}$ — a.s. right lower semi-continuous paths. Consequently, for each $q \geq 1$, ${}^qV_{\cdot \wedge \tau_q}$ is RCLL and so is $y_{\cdot \wedge \tau_q}$. Finally, noticing by the definition of τ_q that

$$\mathbf{P}(\{\omega : \exists q_0(\omega) \geq 1 \text{ such that } \forall q \geq q_0(\omega), \tau_q(\omega) = T\}) = 1, \quad (38)$$

we can conclude that y is also a RCLL process.

Fourth Step: We proceed to prove that $\{(y^n, z^n)\}_{n=1}^\infty$ is a Cauchy sequence in $\mathcal{S}^2(0, T; \mathbf{R}) \times \mathcal{M}^2(0, T; \mathbf{R}^d)$, $\{V^n\}_{n=1}^\infty$, extracting a subsequence if necessary, converges in probability to an increasing continuous process $V \in \mathcal{V}^2(0, T; \mathbf{R})$ uniformly with respect to t , and that Y is a g -supersolution on $[0, T]$.

First, it follows from the pointwise convergence of the sequence $\{y^n\}_{n=1}^\infty$ and Fatou's Lemma as well as (30) that

$$\begin{aligned} 0 \leq \mathbf{E} \left[\int_0^T e^{-s} (Y_s - y_s) ds \right] &\leq \liminf_{n \rightarrow \infty} \mathbf{E} \left[\int_0^T e^{-s} (Y_s - y_s^n) ds \right] \\ &= \liminf_{n \rightarrow \infty} \frac{1}{n} \mathbf{E}[V_T^n] \leq \liminf_{n \rightarrow \infty} \frac{1}{n} (\mathbf{E}[|V_T^n|^2])^{1/2} = 0, \end{aligned}$$

which together with the facts that y is a RCLL process and Y is a continuous process implies that

$$Y_t = y_t, \quad t \in [0, T].$$

Moreover, noticing that $\mathbf{P}$ — a.s., $\{y_t(\omega) - y_t^n(\omega)\}_{n=1}^\infty$ tends decreasingly to 0 for each $t \in [0, T]$, we can deduce by Dini's theorem that when $T < \infty$, $\mathbf{P}$ — a.s., the sequence $\{(y_t(\omega) - y_t^n(\omega))_{t \in [0, T]}\}_{n=1}^\infty$ converges uniformly to 0 on the interval $[0, T]$. When $T = \infty$, since $\mathbf{P}$ — a.s., $y_t^n(\omega) \uparrow y_t(\omega)$ for each $t \in [0, T]$, and $\lim_{t \rightarrow \infty} (y_t(\omega) - y_t^1(\omega)) = y_T(\omega) - y_T^1(\omega) = 0$, it follows that for each $\varepsilon > 0$, there exists a real number $\bar{T} > 0$ depending on ε and ω such that

$$\forall t > \bar{T}, \quad \forall n \geq 1, \quad |y_t(\omega) - y_t^n(\omega)| \leq y_t(\omega) - y_t^1(\omega) \leq \varepsilon. \quad (39)$$

On the other hand, note that the sequence $\{(y_t(\omega) - y_t^n(\omega))_{t \in [0, \bar{T}]}\}_{n=1}^\infty$ converges uniformly to 0 on the interval $[0, \bar{T}]$ due to Dini's theorem. There exists a real number N depending

on ε and ω such that

$$\forall t \in [0, \bar{T}], \quad \forall n > N, \quad |y_t(\omega) - y_t^n(\omega)| \leq \varepsilon.$$

Combining the previous inequality with (39) yields that $\mathbf{P} - \text{a.s.}$, for each $\varepsilon > 0$, if $n > N$, then

$$\sup_{t \in [0, \infty]} |y_t(\omega) - y_t^n(\omega)| \leq \varepsilon,$$

which means that $\mathbf{P} - \text{a.s.}$, $\{(y_t(\omega) - y_t^n(\omega))_{t \in [0, T]}\}_{n=1}^{\infty}$ also converges uniformly to 0 on the interval $[0, T]$ when $T = \infty$. Then, Lebesgue's dominated convergence theorem yields that, no matter $T < \infty$ or $T = \infty$,

$$\lim_{n \rightarrow \infty} \mathbf{E} [|(\gamma - \gamma^n)_T^*|^2] = 0. \quad (40)$$

Next we prove the strong convergence of $\{z^n\}_{n=1}^{\infty}$ in $\mathcal{M}^2(0, T; \mathbf{R}^d)$. For each $m, n \geq 1$, set $\hat{y}^{m,n} := y^m - y^n$, $\hat{z}^{m,n} := z^m - z^n$ and $\hat{V}^{m,n} := V^m - V^n$. Itô's formula yields that for each $t \in [0, T]$ and $m, n \geq 1$,

$$\begin{aligned} |\hat{y}_t^{m,n}|^2 + \int_t^T |\hat{z}_s^{m,n}|^2 ds &= 2 \int_t^T \hat{y}_s^{m,n} (g(s, y_s^m, z_s^m) - g(s, y_s^n, z_s^n)) ds \\ &\quad + 2 \int_t^T \hat{y}_s^{m,n} d\hat{V}_s^{m,n} - 2 \int_t^T \hat{y}_s^{m,n} \langle \hat{z}_s^{m,n}, dB_s \rangle, \end{aligned}$$

from which it follows that

$$\begin{aligned} \mathbf{E} \left[\int_0^T |\hat{z}_s^{m,n}|^2 ds \right] &\leq 2 \mathbf{E} \left[\int_0^T \hat{y}_s^{m,n} (g(s, y_s^m, z_s^m) - g(s, y_s^n, z_s^n)) ds \right] \\ &\quad + \mathbf{E} \left[\int_0^T \hat{y}_s^{m,n} d\hat{V}_s^{m,n} \right]. \end{aligned} \quad (41)$$

The first term on the right hand side in (41) can be enlarged by (H4), (H5), Hölder's inequality and (30) as follows, adding and subtracting the term $g(s, y_s^m, z_s^n)$,

$$\begin{aligned} &\mathbf{E} \left[\int_0^T \hat{y}_s^{m,n} (g(s, y_s^m, z_s^m) - g(s, y_s^n, z_s^n)) ds \right] \\ &\leq \mathbf{E} \left[\int_0^T (u(s) \rho(|\hat{y}_s^{m,n}|^2) + v(s) |\hat{y}_s^{m,n}| |\hat{z}_s^{m,n}|) ds \right] \\ &\leq \mathbf{E} \left[\int_0^T u(s) \rho(|\hat{y}_s^{m,n}|^2) ds \right] + \left(\sup_{m,n \geq 1} \mathbf{E} \left[\int_0^T |\hat{z}_s^{m,n}|^2 ds \right] \right)^{1/2} \\ &\quad \times \left(\mathbf{E} \left[\int_0^T v^2(s) |\hat{y}_s^{m,n}|^2 ds \right] \right)^{1/2}. \end{aligned}$$

Now since $\rho \in \mathbb{S}$, by Jensen's inequality and (40) we can deduce that, as $m, n \rightarrow \infty$,

$$\mathbf{E} \left[\int_0^T u(s) \rho(|\hat{y}_s^{m,n}|^2) ds \right] \leq \int_0^T u(s) ds \cdot \rho \left(\mathbf{E} [|\hat{y}_T^{m,n,*}|^2] \right) \rightarrow 0,$$

and we also have

$$\mathbf{E} \left[\int_0^T v^2(s) |\hat{y}_s^{m,n}|^2 ds \right] \leq \int_0^T v^2(s) ds \cdot \mathbf{E} [|\hat{y}_T^{m,n,*}|^2] \rightarrow 0.$$

Then the first term on the right hand side of (41) tends to 0 when sending m, n to ∞ . Concerning the second term, by Hölder's inequality, (30) and (40) we can get that, as $m, n \rightarrow \infty$,

$$\mathbf{E} \left[\int_0^T \hat{y}_s^{m,n} d\hat{V}_s^{m,n} \right] \leq \left(\sup_{m,n \geq 1} \mathbf{E} [|\hat{V}_T^{m,n}|^2] \right)^{1/2} (\mathbf{E} [|\hat{y}_T^{m,n,*}|^2])^{1/2} \rightarrow 0.$$

Hence, $\{z^n\}_{n=1}^\infty$ is a Cauchy sequence in $\mathcal{M}^2(0, T; \mathbf{R}^d)$, and the limit is denoted by $(z_t)_{t \in [0, T]}$.

In what follows we prove that $\{V^n\}_{n=1}^\infty$, extracting a subsequence if necessary, converges uniformly in probability to an increasing continuous process $V \in \mathcal{V}^2(0, T; \mathbf{R})$. In fact, since $\{z^n\}_{n=1}^\infty$ converges in $\mathcal{M}^2(0, T; \mathbf{R}^d)$, we can assume that $d\mathbf{P} \times dt - \text{a.e.}, z^n \rightarrow z$ and $\mathbf{E} \int_0^T \sup_n |z_s^n|^2 ds < \infty$, extracting a subsequence if necessary, see Lemma 2.5 in Kobylanski [13] for details. Thus, from the continuity properties of the stochastic integral we get that

$$\sup_{t \in [0, T]} \left| \int_0^t \langle z_s^n, dB_s \rangle - \int_0^t \langle z_s, dB_s \rangle \right| \xrightarrow{\mathbf{P}} 0. \quad (42)$$

Moreover, by virtue of (H2), (H5) and the convergence of $\{(y^n, z^n)\}_{n=1}^\infty$, we can have that, as $n \rightarrow \infty$,

$$g(t, y_t^n, z_t^n) \rightarrow g(t, y_t, z_t), \quad d\mathbf{P} \times dt - \text{a.e.} \quad (43)$$

Let the stopping time sequence $\{\tau_q\}_{q=1}^\infty$ be defined in the third step. By (35) we know that $d\mathbf{P} \times dt - \text{a.e.}$, for each $n \geq 1$ and $q \geq 1$,

$$|g(t, y_t^n, z_t^n) \mathbf{1}_{t \leq \tau_q}| \leq \psi_q(t) + |g(t, 0, 0)| + v(t) \sup_{n \geq 1} |z_t^n|.$$

Then Lebesgue's dominated convergence theorem, (43) and (38) contribute to that

$$\sup_{t \in [0, T]} \left| \int_0^t g(s, y_s^n, z_s^n) ds - \int_0^t g(s, y_s, z_s) ds \right| \xrightarrow{\mathbf{P}} 0. \quad (44)$$

Combing (40), (42) and (44), in view of (26), we can conclude that as $n \rightarrow \infty$,

$$V_t^n = y_0^n - y_t^n - \int_0^t g(s, y_s^n, z_s^n) ds + \int_0^t \langle z_s^n, dB_s \rangle$$

uniformly converges in probability to

$$V_t := y_0 - y_t - \int_0^t g(s, y_s, z_s) ds + \int_0^t \langle z_s, dB_s \rangle,$$

i.e. we have that

$$\sup_{t \in [0, T]} |V_t^n - V_t| \xrightarrow{\mathbf{P}} 0. \quad (45)$$

Hence, V is a continuous process, and it is increasing by the first step. Furthermore, in view of (45) and (30), by Fatou's Lemma we know that $V \in \mathcal{V}^2(0, T; \mathbf{R})$.

Finally, taking limit under ucp with respect to n in both sides of BSDE (26) follows that

$$y_t = Y_T + \int_t^T g(s, y_s, z_s) ds + \int_t^T dV_s - \int_t^T \langle z_s, dB_s \rangle, \quad t \in [0, T].$$

That is to say, $Y = y$ is a g -supersolution on $[0, T]$ with terminal condition Y_T , and (z, V) is the unique decomposition of Y on $[0, T]$. This completes the proof of Theorem 18. $\square$

Remark 19: When g satisfies (H3), the monotonic limit theorem for only RCLL g -supersolution on infinite time intervals has not been established. So the continuity condition of g -supermartingales is needed in Theorem 18. It is an open question whether the continuity assumption on V can be omitted.

If the generator g is independent of variable y , we can obtain the Doob-Meyer decomposition for g -supermartingales analogously to the decomposition for classical supermartingales.

Corollary 20: Assume that the generator g is independent of y and satisfies (H1) and (H5). If $Y \in \mathcal{S}^2(0, T; \mathbf{R})$ is a continuous g -supermartingale on $[0, T]$, then it admits the decomposition $Y_t = M_t - V_t$ for each $t \in [0, T]$, where $M \in \mathcal{S}^2(0, T; \mathbf{R})$ is a continuous g -martingale on $[0, T]$ and $V \in \mathcal{V}^2(0, T; \mathbf{R})$ is a continuous increasing process.

Proof: Let assumptions hold. It follows from Theorem 18 that there exists a unique continuous increasing process $V \in \mathcal{V}^2(0, T; \mathbf{R})$ and a process $z \in \mathcal{M}^2(0, T; \mathbf{R}^d)$ satisfying the following BSDE,

$$Y_t = Y_T + \int_t^T g(s, z_s) ds - \int_t^T \langle z_s, dB_s \rangle + V_T - V_t, \quad t \in [0, T].$$

We can set $M_t := Y_t + V_t$, then

$$M_t = Y_T + V_T + \int_t^T g(s, z_s) ds - \int_t^T \langle z_s, dB_s \rangle, \quad t \in [0, T]. \quad (46)$$

By Theorem 6 the pair of $(M_t, z_t)_{t \in [0, T]}$ is a solution of BSDE (46) in $\mathcal{S}^2(0, T; \mathbf{R}) \times \mathcal{M}^2(0, T; \mathbf{R}^d)$. From Definitions 15 and 17 we know that $M \in \mathcal{S}^2(0, T; \mathbf{R})$ is a g -martingale on $[0, T]$. Hence, the proof is finished. $\square$

Remark 21: Accordingly, if the generator g satisfies (H1)–(H5) and the following (H6’):

(H6’) $d\mathbf{P} \times dt$ – a.e., for each $y \in \mathbf{R}$, $g(t, y, 0) \leq g_t + u(t)|y|$, where $(g_t)_{t \in [0, T]}$ and $u(\cdot)$ are defined in (H6),

then a continuous g -submartingale also admits a Doob-Meyer’s decomposition.

Indeed, if g satisfies the above assumptions and $Y \in \mathcal{S}^2(0, T; \mathbf{R})$ is a continuous g -submartingale on $[0, T]$, then $\bar{Y} := -Y$ is a continuous $\bar{g}$ -supermartingale on $[0, T]$, where $\bar{g}(t, y, z) := -g(t, -y, -z)$. Thus, $\bar{Y}$ and $\bar{g}$ satisfy assumptions in Theorem 18, which implies that there exists a unique continuous increasing process $V \in \mathcal{V}^2(0, T; \mathbf{R})$ and a process $\bar{Z} \in \mathcal{M}^2(0, T; \mathbf{R}^d)$ satisfying the following BSDE

$$\bar{Y}_t = \bar{Y}_T + \int_t^T \bar{g}(s, \bar{Y}_s, \bar{Z}_s) ds + \int_t^T dV_s - \int_t^T \langle \bar{Z}_s, dB_s \rangle, \quad t \in [0, T].$$

Then we can get that, setting $Z_t := -\bar{Z}_t$,

$$Y_t = Y_T + \int_t^T g(s, Y_s, Z_s) ds - \int_t^T dV_s - \int_t^T \langle Z_s, dB_s \rangle, \quad t \in [0, T],$$

which is the Doob-Meyer decomposition for the continuous g -submartingale Y on $[0, T]$.

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